

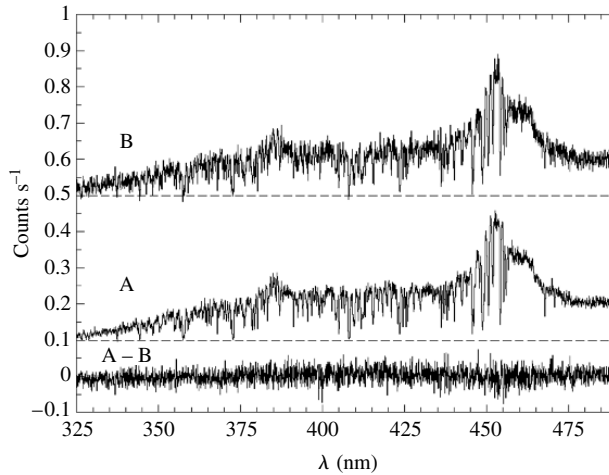


FIGURE 28.34 Spectra of the images of the quasar Q0142–100 formed by a gravitational lens. The bottom panel shows the difference between the two spectra. (Figure adapted from Smette et al., *Ap. J.*, 389, 39, 1992.)

shows a quasar with a redshift of $z = 1.41$. The gravitational lens is due to the gravity of an intervening giant cD galaxy with $z = 0.36$ that is between the two images and $0.8''$ away from one of them. In addition to having the same redshift, both images have the same two bright emission lines and many absorption features in common. Both images also show the same radio core and jet structure.

Figure 28.34 shows spectra of the two images of another quasar, Q0142–100, that are also formed by a gravitational lens. Like optical lenses, gravitational lenses can magnify and increase an object’s brightness (recall Fig. 24.15). The brighter of the two images of Q0142–100 has the appearance of being one of the most luminous quasars known because of the gravitational amplification of the image. (The difference in the apparent magnitudes of the images of Q0142–100 is about $\Delta m_V = 2.12$.)

The Geometry of Gravitational Lensing

Gravitational lensing results when light follows the straightest possible worldline (a geodesic, see Section 17.2) as it travels through the curved spacetime around a massive object. It is analogous to the normal refraction of light by a glass lens that occurs as the light crosses the lens surface, passing from one index of refraction, n , to another, where $n \equiv c/v$ is just the ratio of the speed of light in a vacuum to its speed, v , in the medium. Outside of a spherical object of mass M (which is equivalent to a point mass), the coordinate speed of light in the radial direction is given by Eq. (17.28),

$$\frac{dr}{dt} = c \left(1 - \frac{2GM}{rc^2} \right)$$

so the effective “index of refraction” is

$$n = \frac{c}{dr/dt} = \left(1 - \frac{2GM}{rc^2} \right)^{-1} \simeq 1 + \frac{2GM}{rc^2}$$

for radially traveling light, assuming that $2GM/rc^2 \ll 1$. At a distance of 10^4 pc from a galaxy with a mass of $10^{11} M_\odot$, the effective index of refraction is $n = 1 + 9.6 \times 10^{-7}$. (Of course, the light passing by the point mass will never be traveling exactly radially. This was merely used to estimate the magnitude of the effect of gravity in a gravitational lens.) Obviously, the deviation of the light from a straight line will be extremely small.

Figure 28.35 shows the path taken by light from a source at point S , as it is deflected through an angle, ϕ , by the gravitational lens due to a point mass, M , at point L . The light arrives at the position of the observer at point O .

In Problem 17.6, you found that the angular deviation of a photon passing a distance r_0 (very nearly the distance of closest approach) from a mass M was

$$\phi = \frac{4GM}{r_0 c^2} \text{ rad} \quad (28.20)$$

[this includes the factor of 2 mentioned in part (c) of that problem]. The distance to the source is $d_S / \cos \beta \simeq d_S$, where $\beta \ll 1$, and d_L is the distance to the lensing mass. It is then a matter of simple trigonometry (left as an exercise) to show that the angle θ between the lensing mass and the image of the source must satisfy the equation

$$\theta^2 - \beta\theta - \frac{4GM}{c^2} \left(\frac{d_S - d_L}{d_S d_L} \right) = 0, \quad (28.21)$$

where θ and β are measured in radians.

The quadratic equation (Eq. 28.21) indicates that for the geometry shown in the figure, there will be two solutions for θ , and so two images will be formed by the gravitational lens. Designating these solutions as θ_1 and θ_2 , these angles can be measured observationally and then used to find the values of β and M . The results are

$$\beta = \theta_1 + \theta_2 \quad (28.22)$$

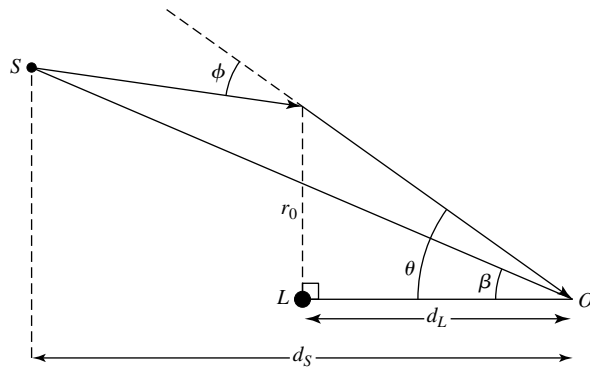


FIGURE 28.35 The geometry for a gravitational lens. Light from the source, S , passes within a distance of approximately r_0 of a lensing point mass at L on its way to an observer at O . The angles involved are actually just a fraction of a degree, and so r_0 is very nearly the distance of closest approach.

and

$$M = -\frac{\theta_1 \theta_2 c^2}{4G} \left(\frac{d_S d_L}{d_S - d_L} \right). \quad (28.23)$$

Referring back to Fig. 28.35, note that Eq. (28.22) implies that θ_1 and θ_2 have opposite signs. As a result, the two images are formed on opposite sides of the gravitational lens, so M will be positive.

Example 28.4.1. For the quasar Q0957+561 shown in Fig. 28.33, $\theta_1 = 5.35'' = 2.59 \times 10^{-5}$ rad, and $\theta_2 = -0.8'' = -3.88 \times 10^{-6}$ rad. (Which angle assumes the minus sign is arbitrary.) From the quasar's redshift of $z_S = 1.41$ and the gravitational lens redshift of $z_L = 0.36$, the Hubble law gives the corresponding distances of $d_S = 2120h^{-1}$ Mpc and $d_L = 890h^{-1}$ Mpc. Then Eq. (28.23) gives

$$M = -\frac{\theta_1 \theta_2 c^2}{4G} \left(\frac{d_S d_L}{d_S - d_L} \right) = 8.1 \times 10^{11} h^{-1} M_\odot.$$

This is in good agreement with a value of $M = 8.7 \times 10^{11} h^{-1} M_\odot$ obtained with a more accurate treatment of the mass distribution of the lensing galaxy.

Einstein Rings and Crosses

If a quasar or other bright source lies exactly along the line of sight to the lensing mass, then it will be imaged as an **Einstein ring** encircling the lens (this phenomenon was described by Einstein in 1936). In this case, $\beta = 0$ in Fig. 28.35, and so Eq. (28.21) can be solved immediately for the angular radius of the Einstein ring,

$$\theta_E = \sqrt{\frac{4GM}{c^2} \left(\frac{d_S - d_L}{d_S d_L} \right)} \text{ rad}. \quad (28.24)$$

Of course, for a point source, the chance of an exact alignment with the lensing mass is essentially zero. For an extended source, the requirements for an Einstein ring are that $\beta < \theta_E$ and that the line of sight through the lensing mass must pierce the extended source. Figure 28.36 shows a calculation of a partial ring—the image of a slightly off-center source. The first Einstein ring to be discovered, MG1131+0456, was found at radio wavelengths by the VLA. Figure 28.37 shows the radio appearance of the ring, which is thought to be the image of a radio galaxy lensed by an elliptical galaxy.

The value of θ_E can be calculated for any gravitational lens, regardless of the alignment of the lens and the source. Although the image may not be a ring, θ_E does provide a useful parameter for describing the properties of any gravitational lens. If $\beta < \theta_E$, as shown in Fig. 28.35, there will be two images formed by the point mass. If $\beta \gg \theta_E$, the position and brightness of the source are only slightly altered, but a secondary image appears close to the lensing mass that is reduced in angular size by a factor of $(\theta_E/\beta)^4$.

A point mass is clearly a crude representation of an actual galaxy. A better model of the lensing galaxy is provided by an isothermal sphere around a central core, similar to the

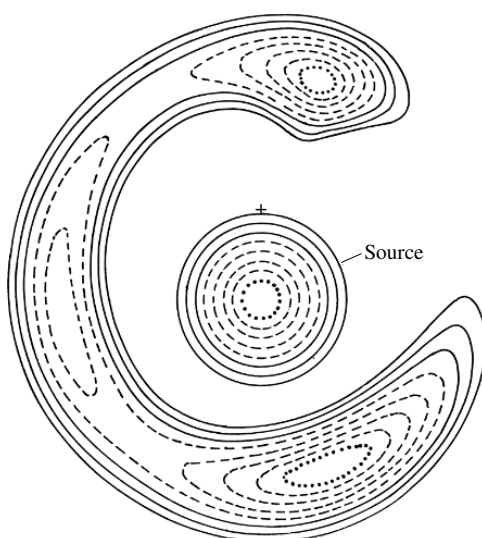


FIGURE 28.36 A calculation of the image of a slightly off-center spherical galaxy formed by a lensing mass located at the cross (“+”). (Figure adapted from Chitre and Narasimha, *Gravitational Lenses*, Springer-Verlag, Berlin, 1989.)

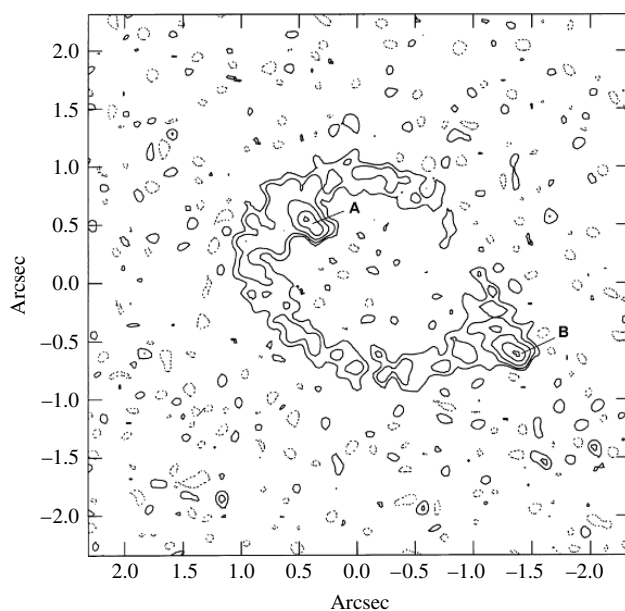


FIGURE 28.37 The Einstein ring MG1131+0456. The knot labeled A is produced by the core of the imaged radio galaxy, and the knot labeled B represents one of its lobes. (Figure adapted from Hewitt et al., *Nature*, 333, 537, 1988. Courtesy of J. Hewitt.)

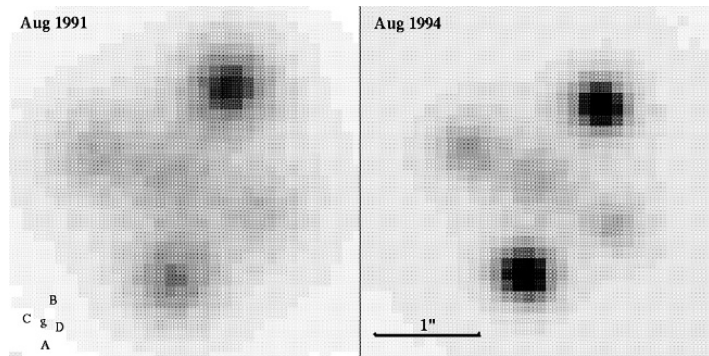


FIGURE 28.38 The Einstein cross Q2237+031, as observed in August 1991 (left) and August 1994 (right). The cross consists of 4 images of the quasar (labeled A–D), with the lensing galaxy (labeled g) at the center. (Courtesy of Geraint Lewis and Mike Irwin.)

model used for the central bulge of the Milky Way; see Section 24.4. Another improvement is to depart from spherical symmetry and use an isothermal ellipsoid, which can produce either three or five images (an extended distribution of mass will produce an odd number of images). The **Einstein cross** shown in Fig. 28.38 includes four images of a distant quasar (Q2237+031, at $z = 1.69$) that is lensed by a nearby ($z = 0.04$) spiral galaxy. There is probably also a fifth faint central image that is overwhelmed by the lensing galaxy at the center of the cross. Note that image A has brightened by 0.5 mag in the 3-year interval between the photos.

Luminous Arcs in Galaxy Clusters

Another striking example of gravitational lensing is the formation of arcs by light passing through a cluster of galaxies. One such arc in the cluster Abell 370 is shown in Fig. 28.39. Up to 60 additional “arclets” and several distorted distant background galaxies have also been observed in that cluster. The source of the large arc must be a resolved object such as a galaxy rather than the starlike nucleus of a quasar. According to one model of Abell 370, the lensing mass (visible galaxies and dark matter) needed to produce the images in Abell 370 is at most about $5 \times 10^{14} M_{\odot}$. Taken with the combined luminosity of a few $\times 10^{11} L_{\odot}$ for the lensing galaxies, this implies a mass-to-light ratio of at least $1000 M_{\odot}/L_{\odot}$, indicating the presence of large amounts of dark matter.

Abell 370 is an unusual cluster in that it is sufficiently centrally condensed to produce such arcs. The dark matter in most clusters is probably more widely distributed, producing weak lensing effects just strong enough to distort the appearance of distant galaxies seen beyond the cluster. Figure 28.40 shows a spectacular example of multiple arclets that are lensed images of background galaxies produced by the cluster Abell 2218. Such weak lensing can also cause an apparent bunching of quasars, so statistical studies of the clustering of objects in the very early universe must take this effect into account.

Time Variability of Multiple Images

An interesting effect occurs when the source for a pair of images increases its luminosity. Because the light from the source takes different paths on its way to the observer, there

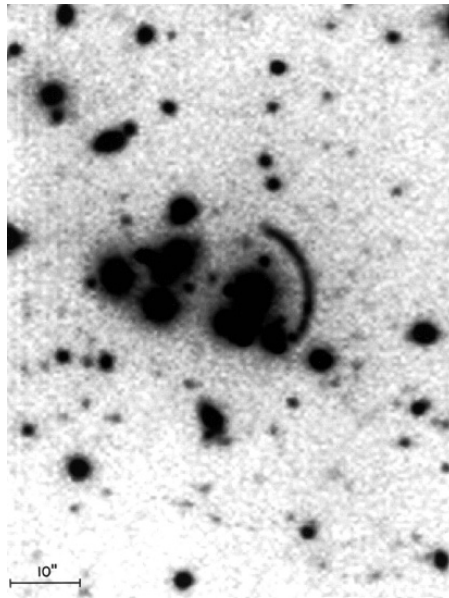


FIGURE 28.39 A gravitationally lensed arc, about $20''$ long, produced by the cluster Abell 370. (Figure from Lynds and Petrosian, *Ap. J.*, 336, 1, 1989. Courtesy of National Optical Astronomy Observatories/R. Lynds.)

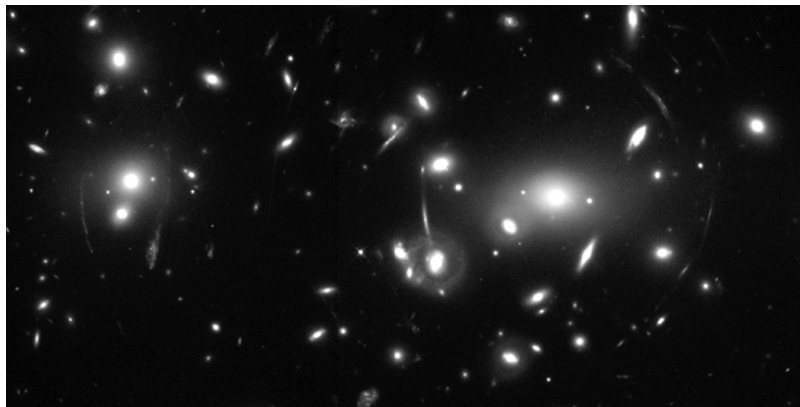


FIGURE 28.40 An HST view of the gravitationally lensed images of background galaxies produced by the cluster Abell 2218. [Courtesy of W. Couch (U. of South Wales), R. Ellis (Cambridge U.), and NASA.]

will be a time delay between the brightening of the lensed images. A time delay of about 1.4–1.5 yr has been measured for the original double quasar, Q0957+561. Nonperiodic celestial events usually catch astronomers by surprise, but this time delay puts astronomers in the unique situation of knowing in advance how a lensed quasar will behave.

It turns out that the time delay is also inversely proportional to the Hubble constant. This offers a way of determining the value of H_0 that is independent of any other distance

measurement. At the cosmological distances of quasars, their recessional velocities should completely dominate their peculiar velocities through space. One study using Q0957+561 concluded that $H_0 = 69 \pm 21 \text{ km s}^{-1} \text{ Mpc}^{-1}$, assuming that the lensing galaxy contains a substantial amount of dark matter. The result is in excellent agreement with the WMAP value of $H_0 = 71_{-3}^{+4} \text{ km s}^{-1} \text{ Mpc}^{-1}$.

The Lyman-Alpha Forest

The spectra of high-redshift quasars always display a large number of narrow absorption lines superimposed on the quasar's continuous spectrum (these lines are in addition to any broad absorption lines that are associated with the quasar itself). These narrow lines are formed when the light from a quasar passes through material (an interstellar cloud, a galactic halo) that happens to lie along the line of sight. If the absorbing material is far from Earth, its recessional motion will cause these absorption lines to be strongly redshifted. Furthermore, if the light passes through more than one cloud or galactic halo during its trip to Earth, different sets of absorption lines will be seen. Each set of lines corresponds to the redshift of a particular cloud or halo.

There are two classes of narrow absorption lines in quasar spectra:

- The **Lyman- α forest** is a dense thicket of hydrogen absorption lines. These lines are believed to be formed in intergalactic clouds and display a variety of redshifts. Absorption by primordial ionized helium (He II) has also been detected.
- Lines are also formed by **ionized metals**, primarily carbon (C IV) and magnesium (Mg II), together with silicon, iron, aluminum, nitrogen, and oxygen. The mix of elements is similar to that found in the interstellar medium of the Milky Way, indicating that the material has been processed through stars and enriched in heavy elements. These lines are thought to be formed in the extended halos or disks of galaxies found along the line of sight to the quasar.

Most of these lines are normally found at ultraviolet wavelengths, when the absorbing material is moving at a small fraction of the speed of light relative to Earth (i.e., has a small redshift). They are rarely seen from the ground because Earth's atmosphere absorbs most ultraviolet wavelengths. However, if the absorbing material is receding fast enough, the Doppler effect can shift ultraviolet lines to visible wavelengths, where the atmosphere is transparent. For this reason, these absorption lines are seen from the ground only in the spectra of highly redshifted quasars.

Example 28.4.2. The rest wavelength of the ultraviolet Lyman- α line of hydrogen is $\lambda_{\text{Ly}\alpha} = 121.6 \text{ nm}$ (Section 5.3). To determine the redshift required to bring this line into the visible region of the electromagnetic spectrum, we can use the definition of z ,

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}.$$

continued

Using $\lambda_{\text{rest}} = \lambda_{\text{Ly}\alpha}$ and $\lambda_{\text{obs}} = 400 \text{ nm}$ for the blue end of the visible spectrum, we require a redshift of

$$z > \frac{400 \text{ nm} - 121.6 \text{ nm}}{121.6 \text{ nm}} \simeq 2.3,$$

just to bring the Ly α line to the edge of the visible spectrum. Actually, some near-ultraviolet light can penetrate Earth's atmosphere, so the Ly α line can be observed when $z > 1.7$ for the absorbing material.

Typically, the spectrum of a high-redshift quasar contains a strong Lyman- α emission line produced by the quasar itself, and perhaps some 50 Ly α absorption lines at shorter wavelengths (smaller redshifts); see Fig. 28.41. Each one of these lines is from a different intergalactic cloud of hydrogen (and presumably helium) encountered by the quasar's continuum radiation on its journey to Earth. As explained in Section 9.5, the Ly α line profile can be used to calculate the column density of the neutral hydrogen atoms in the cloud that produces each line. A typical result is 10^{18} m^{-2} . In other words, a hollow tube having a cross-sectional area of 1 m^2 that crossed completely through the cloud would contain 10^{18} neutral hydrogen atoms. Such a cloud would be extremely transparent to the ultraviolet radiation that is normally present throughout space. As a result, this ultraviolet background can penetrate the cloud and keep it almost completely ionized. Calculations indicate that only one hydrogen atom in 10^5 remains neutral in the cloud and is capable of absorbing an ultraviolet photon.

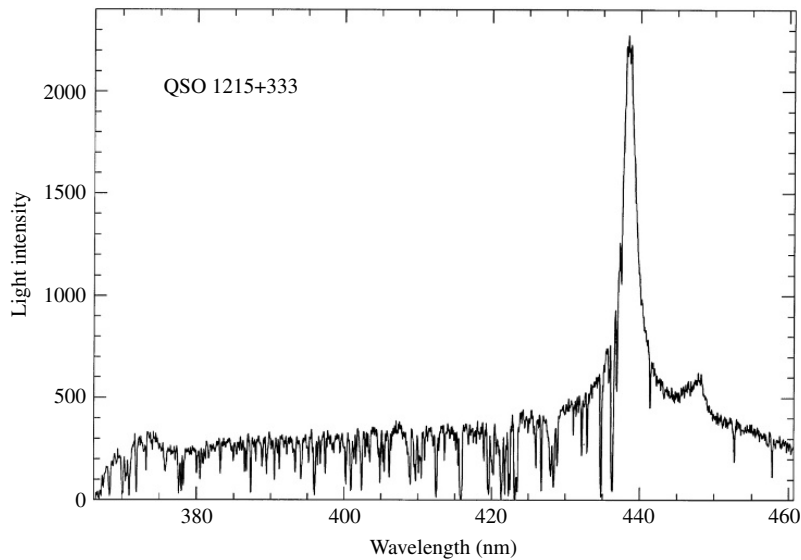


FIGURE 28.41 The strong Ly α emission line in the spectrum of QSO 1215+333, with the Ly α forest of absorption lines at shorter wavelengths. (Adapted from a figure courtesy of J. Bechtold, Steward Observatory, University of Arizona.)

We deduce the size of the intergalactic clouds by comparing the Ly α forest in the spectra of pairs of lensed quasars. Many of the absorption lines are seen in both spectra, but some are not. This indicates that the clouds are, on average, about the size of the lensing galaxy. From the total calculated column density of hydrogen (ionized as well as neutral), the mass of a typical cloud probably lies between 10^7 and 10^8 $M_{\odot}$. At the temperature estimated for a typical cloud by some astronomers (approximately 3×10^4 K), its self-gravity would be too weak to keep it from dispersing. It may be held together by the pressure of a less dense (but hotter) external intergalactic medium or by the presence of dark matter within the cloud.

Ionized Metal Absorption Lines in Quasars

The narrow absorption lines produced by ionized metals in quasar spectra have a different origin. They can be divided into two groups as observed from Earth's surface, corresponding to two different redshift ranges. Below roughly $z = 1.5$, the Mg II lines dominate, accompanied by Si II, C II, Fe II, and Al II, because they fall within the wavelength window that can be seen from the ground (the Mg II lines are probably produced in the halos of normal galaxies or in regions of star formation.). The C IV lines, together with Si IV, N V, and O IV, are common between about $z = 1.2$ and $z = 3.5$, however.

The distribution of redshifts of these lines is in general agreement with the expected distribution of galaxies at that earlier time when the universe was smaller by a factor of $1 + z$, assuming that the galactic halos are typically some 30–50 kpc across. In fact, some Mg II systems with $z < 1$ have been clearly identified with foreground galaxies seen in direct images. The C IV lines probably come from clouds in young galaxies that are strongly ionized by young, hot OB stars. These narrow metal lines indicate lower-than-solar abundances of heavy elements, consistent with their origin in young galaxies that may still be in the process of forming.

The Density Distribution of Intergalactic Clouds

The comoving space density of intergalactic clouds appears to have been greater in the past than it is today, so the number of clouds has been decreasing as the universe ages. A statistical analysis of the clouds' redshifts reveals little evidence that the clouds tend to be grouped in clusters. Instead, they appear to be distributed randomly throughout space. In particular, there do not appear to be large voids in the distribution of these intergalactic clouds similar to those described in Section 27.3 for galaxies (the significance of this is not yet clear). The distribution of the He II is similarly uncertain.

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PROBLEMS

- 28.1** The radio galaxy Centaurus A has a redshift of $z = 0.00157$. The monochromatic flux of Cen A is $F_\nu = 912$ Jy at a frequency of 1400 MHz. Using $\alpha = 0.6$ for its spectral index, estimate the radio luminosity of Cen A.
- 28.2** Use Fig. 28.14 to calculate the spectral index, α , of the quasar 3C 273 at a radio frequency of 1400 MHz. Compare your answer with the value given in Example 28.1.2.
- 28.3** For a temperature of 7.3×10^5 K, make a graph of the Planck function (Eq. 3.24), plotting $\log_{10} \nu B_\nu(T)$ vs. $\log_{10} \nu$ for $\log_{10} \nu$ between 15.5 and 17.5. How does the behavior of your graph of a blackbody compare with that of Fig. 28.14 for the continuous spectrum of the quasar 3C 273?
- 28.4** A rough idea of how the population of quasars may have been different in the past can be gained by mathematically modeling the dimming of quasars as they age.
- (a) Consider the case where the total number of quasars has remained constant back to $z = 2.2$, and suppose that the average luminosity, L , of a quasar with a redshift z has the form

$$L = L_0(1 + z)^a,$$

where L_0 is the luminosity at $z = 0$ (today). Use Fig. 28.16 to estimate the value of the constant a .

- (b) From your answer to part (a), how much more luminous is an average quasar at $z = 2$ than at $z = 0$?
- 28.5** Use Newtonian physics to calculate the values of the average density and “surface gravity” for a $10^8 M_\odot$ black hole. Compare these values with those for the Sun.
- 28.6** Use the Newtonian expression for the disk luminosity, Eq. (18.23), to estimate the efficiency of the accretion luminosity of a disk around a nonrotating black hole ($R = 3R_S$). Repeat this for a maximally rotating black hole ($R = 0.5R_S$).
- 28.7** The maximum possible angular momentum for an electrically neutral rotating black hole is

$$L_{\max} = \frac{GM^2}{c}$$

(see Problem 17.19). Use Newtonian physics to make estimates for this problem.

- (a) What is the maximum angular velocity, $\omega_{\max}$, for a $M = 10^8 M_\odot$ black hole? Use MR_S^2 as an estimate of the black hole’s moment of inertia, where R_S is the Schwarzschild radius.
- (b) Consider a straight wire with a length $\ell = R_S$ that rotates about one end with angular velocity $\omega_{\max}$ perpendicular to a uniform magnetic field of $B = 1$ T. What is the induced voltage between the ends of the wire?
- (c) If a battery with the voltage found in part (b) were connected to a wire with a resistance of 30Ω , how much power would be dissipated by the wire?

28.8 Repeat the derivation of Eq. (12.32) for the Strömgren radius, including the effect of the ionized gas filling only a fraction, ϵ , of the total volume, and so arrive at Eq. (28.13) for the radius of the narrow-line region.

28.9 Use the values in Example 28.2.2 to find the fraction of all photons in the continuum (frequencies between 10^{10} Hz and 10^{25} Hz) that are capable of ionizing hydrogen in the ground state.

28.10 Starting with Eq. (28.15) for superluminal motion, derive Eqs. (28.16), (28.17), (28.18), and (28.19).

28.11 Consider material that is ejected from a quasar directly toward Earth.

(a) If the redshift of the quasar is z_Q and the redshift of the ejecta is z_{ej} , show that the speed of the ejecta relative to the quasar is given by

$$\frac{v}{c} = \frac{(1 + z_Q)^2 - (1 + z_{ej})^2}{(1 + z_Q)^2 + (1 + z_{ej})^2}.$$

(b) Consider a radio-emitting knot ejected from the quasar 3C 273 directly toward Earth. If astronomers measure the speed of the approaching knot at $v = 0.9842c$, what is the speed of the knot relative to the quasar? From the frame of reference of the quasar, what is the value of the knot's Lorentz factor?

28.12 Consider a relativistic ($\gamma \gg 1$) blazar jet that is coming directly toward the observer. If there is a time variation Δt_{rest} in the rest frame of the jet, use Eq. (4.31) to show that the variation observed at Earth is approximately

$$\Delta t_{\text{obs}} \simeq \frac{\Delta t_{\text{rest}}}{2\gamma}.$$

28.13 Estimate the effective index of refraction for light passing within 10^4 pc of a spherical cluster of galaxies of a total mass of $10^{14} M_\odot$.

28.14 Verify that Eq. (28.20) gives the correct numerical value for the angular deflection of a light ray that grazes the Sun's surface.

28.15 Use the geometry of the gravitational lens shown in Fig. 28.35 and the value of ϕ given by Eq. (28.20) to derive Eqs. (28.21), (28.22), and (28.23). *Hint:* Start by showing that

$$\frac{\sin(\theta - \beta)}{d_s - d_L} = \frac{\sin \phi}{d_s}$$

for the small angles involved in this problem.

28.16 Two images of the quasar Q0142–100 (also known as UM 673) are formed by a gravitational lens. The redshift of the quasar is $z = 2.727$, and the redshift of the imaging galaxy is $z = 0.493$. The images are separated by $2.22''$, and the lensing galaxy is along the line between the two images, $0.8''$ from one of them. Estimate the mass of the lensing galaxy.

28.17 The Einstein ring MG1654+1346 has a diameter of $2.1''$. The redshift of the source is $z = 1.74$, and the redshift of the lensing galaxy is $z = 0.25$. Estimate the mass of the lensing galaxy.

28.18 In Section 24.2, the detection of a MACHO by its gravitational microlensing of a star in the Large Magellanic Cloud was described. Assume that a MACHO with a mass ten times that of Jupiter orbits halfway between Earth and the LMC and that it moves perpendicular to our line of sight to the LMC. How much time will it take for the MACHO to move through an angle

of $2\theta_E$ across a lensed star in the LMC? Take the distance to the LMC to be 52 kpc and the orbital velocity of the MACHO to be 220 km s^{-1} . Neglect the motion of Earth and the LMC in this problem. Comment on a comparison of your answer with the time shown in Fig. 24.15.

- 28.19** When a small object approaches a much more massive object, the smaller object can be tidally disrupted. The distance of closest approach before being tidally disrupted is the Roche limit (see Eq. 19.4). If the small object is a star and the large object is a supermassive black hole, the Roche limit is given by

$$r_R = 2.4 \left(\frac{\bar{\rho}_{\text{BH}}}{\bar{\rho}_*} \right)^{1/3} R_S,$$

where R_S is the Schwarzschild radius, $\bar{\rho}_{\text{BH}}$ is the density of the black hole, and $\bar{\rho}_*$ is the average density of the star.

- Setting the average density of the supermassive black hole equal to its mass divided by the volume contained within the Schwarzschild radius, derive an expression for the mass of a black hole that would have $r_R = R_S$.
- If the Sun were to fall into a supermassive black hole, what maximum mass could the black hole have if the Sun would be tidally disrupted before crossing the event horizon? Compare your answer to the mass estimates of typical supermassive black holes in galactic nuclei.
- If the supermassive black hole exceeded the mass found in part (b), what would be the implications in terms of liberating the gravitational potential energy of the infalling star? Could infalling stars effectively power AGNs in this case?

COMPUTER PROBLEM

- 28.20** Table 28.2 gives values of the monochromatic flux for Cygnus A at several radio wavelengths.
- Make a graph of $\log_{10} F_\nu$ vs. $\log_{10} \nu$, and determine the value of the spectral index at $\log_{10} \nu = 8$ for the power law given by Eq. (28.1).
 - Use a simple trapezoid rule to integrate the area under the curve of F_ν vs. ν (*not* $\log_{10} F_\nu$ vs. $\log_{10} \nu$!) for the data given, and use your answer to estimate the radio luminosity of Cygnus A.

TABLE 28.2 Cygnus A Data for Problem 28.20.

$\log_{10} \nu$ (Hz)	$\log_{10} F_\nu$ ($\text{W m}^{-2} \text{ Hz}^{-1}$)
7.0	-21.88
7.3	-21.55
7.7	-21.67
8.0	-21.86
8.3	-22.09
8.7	-22.38
9.0	-22.63
9.3	-22.96
9.7	-23.43
10.0	-23.79

CHAPTER

29

Cosmology

- 29.1 *Newtonian Cosmology*
- 29.2 *The Cosmic Microwave Background*
- 29.3 *Relativistic Cosmology*
- 29.4 *Observational Cosmology*

29.1 ■ NEWTONIAN COSMOLOGY

On December 27, 1831, a ship sailed out of Plymouth, England, on a voyage around the world that would last nearly five years. Only 90 feet long, the *Beagle* was crowded with 74 people, one of whom was Charles Darwin. During stops in South America, the Galapagos Islands, Tahiti, New Zealand, and Australia, he exercised his formidable powers of observation. In 1859, after two decades of careful study and reflection, Darwin published *On the Origin of Species by Means of Natural Selection, or the Preservation of Favored Races in the Struggle for Life*. For the first time, people began to comprehend their own origins.

Other discoveries followed during the next one hundred years, with careful observations and brilliant deductions uncovering more about our beginnings. The elucidation of DNA and plate tectonics revealed the mechanisms by which we and our planet evolved. The ideas of stellar nucleosynthesis explained the manufacture of the chemical elements by stars, implying the origin of our corporeal bodies and the ground on which we walk. Even the universe itself was found to be expanding. Then, in 1964, two researchers at Bell Laboratories measured the afterglow of the Big Bang, confirming the explosive origin of everything in existence. It is difficult to imagine a more breathtaking leap from ignorance to self-knowledge than that which occurred during this century of discovery.

Cosmology, taken as a whole, is the study of the origin and evolution of the universe. In this chapter, cosmology will be considered from several different perspectives. To help develop our intuition, this section will discuss the expansion of the universe from the point of view of Newtonian mechanics, without the complications (or insights) provided by general relativity or the modern ideas of particle physics. The discovery and implications of the cosmic microwave background radiation are described in Section 29.2, followed, in Section 29.3, by an introduction to the geometry of the universe as explained by general relativity. Section 29.4 describes how some of the key parameters of cosmology may be measured observationally. The intriguing theories and speculations provided by particle physics will be reserved for Chapter 30.

Olbers's Paradox

Newton believed in an infinite static universe filled with a uniform scattering of stars. If the distribution of matter did not extend forever, he realized, then it would collapse inward due to its own self-gravity. However, Newton's contemporary, Edmund Halley, worried about a sky filled with an infinite number of stars. Why then, asked Halley, is the sky dark at night?

This question was put in its strongest form by a German physician, Heinrich Olbers (1758–1840). Olbers argued in 1823 that if we live in an infinite, transparent universe filled with stars, then in any direction one looks in the night sky, one's line of sight will fall on the surface of a star. (Similarly, if standing in an infinite forest, in every direction you look you will see a tree.) This conclusion is valid regardless of whether the stars are uniformly distributed, as Newton believed, or grouped in galaxies. Olbers's argument was so strong that its disagreement with the obvious fact that the night sky is indeed dark became known as **Olbers's paradox**.¹

Olbers believed that the answer to this paradox was that space is not transparent. The ideas of thermodynamics were still being developed at that time, and Olbers could not appreciate that his explanation was incorrect. The flaw was that any obscuring matter hiding the stars beyond would be heated up by the starlight until it glowed as brightly as a stellar surface. Surprisingly, the first essentially correct answer came from American poet and author Edgar Allan Poe (1809–1849). Poe proposed that because light has a finite speed and the universe is not infinitely old, the light from the most distant sources has not yet arrived. This solution was independently put on a firm scientific foundation by William Thomson (Lord Kelvin, 1824–1907). In more modern terms, the solution to Olbers's paradox is that our universe is simply too young for it to be filled with light.²

Developing Our Intuition: A Newtonian Approach

You will soon discover that this chapter contains more mathematics than the preceding material.³ The reward for the necessary effort is a *quantitative* description of how the universe unfolded that is much more satisfying than qualitative storytelling. Although general relativity is required for a complete understanding of the structure and evolution of the universe, it is useful to develop some intuition by first considering the expansion of the universe from a Newtonian point of view.

The Cosmological Principle

In Section 27.2 it was argued that Hubble's law, Eq. (27.6), is a natural outcome of an expanding universe that is both *isotropic* and *homogeneous*, appearing the same in all directions and at all locations. This crucial assumption of an isotropic and homogeneous universe is called the **cosmological principle**. To show that the expansion of the universe

¹You should recall the earlier discussion of Olbers's paradox in Section 24.1.

²It is sometimes argued that the cosmological redshift caused by the expansion of the universe is responsible for the darkness of the night sky because it shifts starlight out of the visible spectrum. In fact, this effect is much too small to contribute significantly to a dark night sky.

³In some derivations, many of the intermediate steps have been omitted. Filling in these mathematical gaps has been relegated to several of the end-of-chapter problems.

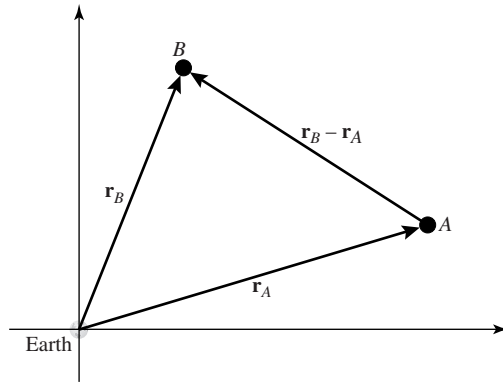


FIGURE 29.1 The expansion of the universe, with Earth at the origin.

appears the same to all observers at all locations, let Earth be at the origin of the coordinate system shown in Fig. 29.1, and consider two galaxies, A and B , located at positions $\mathbf{r}_A$ and $\mathbf{r}_B$, respectively. According to the Hubble law, the recessional velocities of the two galaxies are described by the vectors

$$\mathbf{v}_A = H_0 \mathbf{r}_A$$

and

$$\mathbf{v}_B = H_0 \mathbf{r}_B.$$

The recessional velocity of Galaxy B as seen by an observer in Galaxy A is therefore

$$\mathbf{v}_B - \mathbf{v}_A = H_0 \mathbf{r}_B - H_0 \mathbf{r}_A = H_0 (\mathbf{r}_B - \mathbf{r}_A),$$

so the observer in Galaxy A sees all of the other galaxies in the universe moving away with recessional velocities described by the *same* Hubble law as on Earth.

Although the value of the Hubble “constant,” H_0 , is assumed to be the same for all observers, it is actually a function of time, $H(t)$. If the present time is t_0 , then $H_0 \equiv H(t_0)$.

A Simple Pressureless “Dust” Model of the Universe

To develop an understanding of how the expansion of the universe varies with time, imagine a universe filled with a pressureless “dust” of uniform density, $\rho(t)$, and choose an arbitrary point for the origin. Unlike the actual universe, this model universe is both perfectly isotropic and homogeneous at all scales. The pressureless dust represents all of the matter in the universe after being homogenized and uniformly dispersed. It should not be confused with the physical dust grains found throughout the interstellar medium. There are no photons or neutrinos in this single-component model of the universe.

As the universe expands, the dust is carried radially outward from the origin. Let $r(t)$ be the radius of a thin spherical shell of mass m at time t ; see Fig. 29.2. This shell of mass

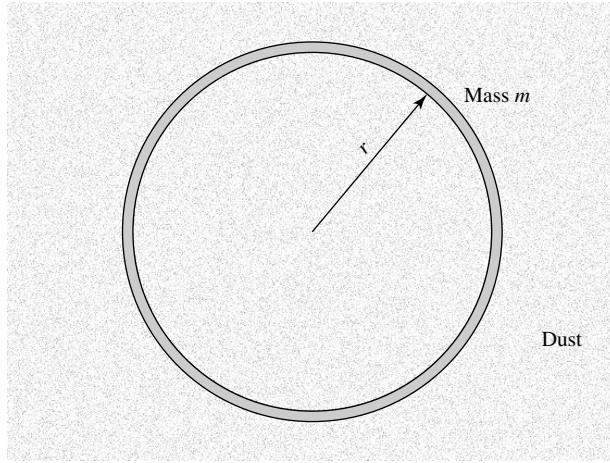


FIGURE 29.2 Spherical mass shell in a dust-filled universe.

expands along with the universe with recessional velocity $v(t) = dr(t)/dt$, so it always contains the same dust particles. Then the mechanical energy E of the shell is

$$K(t) + U(t) = E.$$

As the shell expands, the gravitational pull from the mass inside causes the kinetic energy, K , to decrease while the gravitational potential energy, U , increases.⁴ However, by conservation of energy, the total energy, E , of the shell does not change as the shell moves outward. For future convenience, the total energy of the shell is written in terms of two constants, k and ϖ , such that $E = -\frac{1}{2}mkc^2\varpi^2$. The constant k has units of $(\text{length})^{-2}$; its meaning will be explored in Section 29.3. The other constant, ϖ (“varpi”), labels this particular mass shell and may be thought of as the *present* radius of the shell, so $r(t_0) = \varpi$. The statement of the conservation of the mass shell’s energy is then

$$\frac{1}{2}mv^2(t) - G\frac{M_r m}{r(t)} = -\frac{1}{2}mkc^2\varpi^2. \quad (29.1)$$

In the left-hand side of Eq. (29.1), M_r is the mass interior to the shell,

$$M_r = \frac{4}{3}\pi r^3(t)\rho(t).$$

Although the radius of the shell and the density of the dust are continually changing, the combination $r^3(t)\rho(t)$ does not vary because the mass interior to a specific shell remains constant as the universe expands. Canceling m and substituting for M_r in Eq. (29.1) gives

$$v^2 - \frac{8}{3}\pi G\rho r^2 = -kc^2\varpi^2. \quad (29.2)$$

⁴Recall from Section 10.1 that the mass outside the shell does not contribute to the gravitational force on the shell.

The constant k determines the ultimate fate of the universe:

- If $k > 0$, the total energy of the shell is negative, and the universe is *bounded*, or **closed**. In this case, the expansion will someday halt and reverse itself.
- If $k < 0$, the total energy of the shell is positive, and the universe is *unbounded*, or **open**. In this case, the expansion will continue forever.
- If $k = 0$, the total energy of the shell is zero, and the universe is **flat**, neither open nor closed. In this case, the expansion will continue to slow down, coming to a halt only as $t \rightarrow \infty$ and the universe is infinitely dispersed.

The Newtonian cosmology of this section always takes place in a flat spacetime, with a metric given by Eq. (17.19). The terms *closed*, *open*, and *flat* above should be understood as describing the *dynamics* of the universal expansion. In Section 29.3, these terms will be reinterpreted to describe the geometry of spacetime.

The cosmological principle requires that the expansion proceed in the same way for all shells; for example, the time required for every shell to double its distance from the origin is assumed to be the same. This means that the radius of a particular shell (identified by ϖ) at any time can be written as

$$r(t) = R(t)\varpi. \quad (29.3)$$

In this expression, $r(t)$ is called the **coordinate distance**. Because ϖ labels a shell and follows it as it expands, ϖ is referred to as a **comoving coordinate**; see Fig. 29.3. $R(t)$ is a dimensionless **scale factor** (the same for all shells) that describes the expansion; $R(t_0) = 1$ corresponds to $r(t_0) = \varpi$. The scale factor R is equal to $R_{\text{emitted}}/R_{\text{obs}}$ in Eq. (28.3). Thus

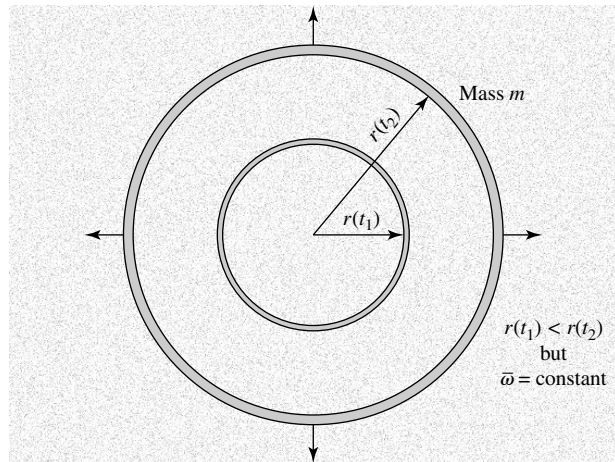


FIGURE 29.3 An expanding mass shell seen at two different times, $t_1 < t_2$. As the mass shell expands, its comoving coordinate, ϖ , is the same at times t_1 and t_2 , while $r(t_1) < r(t_2)$.

R and the redshift z are related by

$$R = \frac{1}{1+z}, \quad (29.4)$$

For example, looking back to a redshift of $z = 3$ implies a universe for which the scale factor was $R = 1/4$.

The previous statement that $r^3 \rho$ does not vary for a specific shell means that $R^3 \rho$ remains constant for *all* shells. That is, since $R(t_0) = 1$,

$$R^3(t)\rho(t) = R^3(t_0)\rho(t_0) = \rho_0, \quad (29.5)$$

where ρ_0 is the density of the dust-filled universe at the present time. Using Eq. (29.4), this can also be written as

$$\rho(z) = \rho_0(1+z)^3, \quad (29.6)$$

which gives the average density of the universe as observed at redshift z . You are cautioned that Eqs. (29.5) and (29.6) are valid only for a universe consisting of pressureless dust. The more general counterpart of Eq. (29.5) will be derived at the end of this section.

The Evolution of the Pressureless "Dust" Universe

The evolution of our Newtonian universe, which can be described by the time behavior of the scale factor $R(t)$, will be investigated in the next few pages. A brief flurry of mathematics will produce the needed tools. The first step is to write the Hubble parameter, $H(t)$, in terms of the scale factor. The Hubble law is

$$v(t) = H(t)r(t) = H(t)R(t)\varpi. \quad (29.7)$$

Because $v(t)$ is the time derivative of $r(t)$, Eq. (29.3) gives

$$v(t) = \frac{dR(t)}{dt} \varpi.$$

Comparing this with Eq. (29.7) shows that

$$H(t) = \frac{1}{R(t)} \frac{dR(t)}{dt}. \quad (29.8)$$

Inserting Eqs. (29.3) and (29.7) into Eq. (29.2) and canceling the ϖ^2 results in

$$\left(H^2 - \frac{8}{3}\pi G\rho \right) R^2 = -kc^2, \quad (29.9)$$

or, employing Eq. (29.8),

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3}\pi G\rho \right] R^2 = -kc^2. \quad (29.10)$$

The left-hand sides of Eqs. (29.9) and (29.10) apply to *all shells* and involve the functions of time $H(t)$, $\rho(t)$, and $R(t)$, while the right-hand sides are constant (the same for all positions and times). Using Eq. (29.5), Eq. (29.10) can be written in terms of R and t only:

$$\left(\frac{dR}{dt}\right)^2 - \frac{8\pi G\rho_0}{3R} = -kc^2. \quad (29.11)$$

This result, along with Eqs. (29.9) and (29.10), will be used to describe the expansion of the universe.

Now we are ready to examine the motion of mass shells in the three cases of a flat, closed, or open universe. First, consider the case of a flat universe ($k = 0$), corresponding to each shell expanding at exactly its escape velocity. The value of the density that will result in a value of $k = 0$ is known as the **critical density**, $\rho_c(t)$. From Eq. (29.9),

$$\rho_c(t) = \frac{3H^2(t)}{8\pi G}. \quad (29.12)$$

To evaluate this at the present time, it is useful to recall (Eq. 27.11) that the Hubble constant in conventional units is

$$H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1} = 3.24 \times 10^{-18} h \text{ s}^{-1} \quad (29.13)$$

which, using the WMAP⁵ value of

$$[h]_{\text{WMAP}} = 0.71,$$

is

$$[H_0]_{\text{WMAP}} = 71 \text{ km s}^{-1} \text{ Mpc}^{-1} = 2.30 \times 10^{-18} \text{ s}^{-1}. \quad (29.14)$$

The present value of the critical density, $\rho_{c,0}$, is then

$$\begin{aligned} \rho_{c,0} &= \frac{3H_0^2}{8\pi G} \\ &= 1.88 \times 10^{-26} h^2 \text{ kg m}^{-3}, \end{aligned} \quad (29.15)$$

with a WMAP value of

$$\rho_{c,0} = 9.47 \times 10^{-27} \text{ kg m}^{-3}. \quad (29.16)$$

This is equivalent to about six hydrogen atoms per cubic meter. However, the WMAP value of the average density of **baryonic matter** in the universe is about 4% of the critical density,

$$\rho_{b,0} = 4.17 \times 10^{-28} \text{ kg m}^{-3} \quad (\text{for } h = 0.71), \quad (29.17)$$

⁵The WMAP (Wilkinson Microwave Anisotropy Probe) measurements of cosmological parameters were introduced in Section 27.2 and will be more fully described in Section 30.2.

or 1 hydrogen atom per 4 m^3 of space. By “baryonic matter,” we mean matter made of *baryons* (e.g., protons and neutrons); hence the “*b*” subscript designating baryonic matter in Eq. (29.17).⁶ As will be explained in Section 29.2, this value is consistent with that obtained from comparing the theoretical and observed abundances of light elements, such as helium-3 and lithium-7, that were formed in the early universe. The density of nonbaryonic dark matter, which is of unknown composition, is not included in the value of $\rho_{b,0}$. Nonbaryonic dark matter is revealed only by its gravitational influence on baryonic matter. Presumably it interacts very weakly (if at all) with photons and charged particles via the electromagnetic force, so it does not absorb, emit, or scatter appreciable amounts of light. Our model universe of pressureless dust includes both types of matter, baryonic and nonbaryonic, luminous and dark.

The ratio of a measured density to the critical density is an important parameter in cosmology. Accordingly, it is useful to define the **density parameter**,

$$\Omega(t) \equiv \frac{\rho(t)}{\rho_c(t)} = \frac{8\pi G\rho(t)}{3H^2(t)}, \quad (29.18)$$

which has a present value of

$$\Omega_0 = \frac{\rho_0}{\rho_{c,0}} = \frac{8\pi G\rho_0}{3H_0^2}. \quad (29.19)$$

Table 29.1 shows the mass-to-light ratios of a variety of astronomical systems and the density parameters derived for them. With the exception of Big Bang nucleosynthesis, these values were obtained by studying gravitational effects and thus include both baryonic and dark matter. There is a significant trend that more extensive systems have larger mass-to-light ratios and density parameters, but, as shown in Fig. 29.4, for the largest systems the density parameters seem to reach a “ceiling” at a maximum value of $\Omega_0 \simeq 0.3$. This is consistent with the WMAP result for the value of the average density of all types of matter, baryonic and dark:

$$[\Omega_{m,0}]_{\text{WMAP}} = (0.135^{+0.008}_{-0.009})h^{-2} = 0.27 \pm 0.04 \quad (\text{for } h = 0.71). \quad (29.20)$$

This corresponds to a mass density of

$$\rho_{m,0} = 2.56 \times 10^{-27} \text{ kg m}^{-3} \quad (\text{for } h = 0.71) \quad (29.21)$$

The “*m*” subscript, which stands for “mass,” anticipates models of the universe with more than one component. This subscript will be suppressed for the present one-component model.

⁶Actually, anything made of known particles whose density obeys Eq. (29.5) counts as baryonic matter. This excludes photons and neutrinos because, as we shall see in Section 29.2, a gas of either of these particles does not obey Eq. (29.5). Such a gas exerts a pressure and therefore is inconsistent with our model universe of pressureless dust.

TABLE 29.1 Mass-to-Light Ratios and Density Parameters, Measured for a Variety of Systems. The complicated dependence on h for the values from the X-ray halo of M87 and Local Group timing is not shown. (Adapted from Binney and Tremaine, *Galactic Dynamics*, Princeton University Press, Princeton, NJ, 1987, and Schramm, *Physica Scripta*, T36, 22, 1991.)

Method	M/L ($M_{\odot}/L_{\odot}$)	Ω_0
Solar neighborhood	3	$0.002h^{-1}$
Elliptical galaxy cores	$12h$	0.007
Local escape speed	30	$0.018h^{-1}$
Satellite galaxies	30	$0.018h^{-1}$
Magellanic Stream	> 80	$> 0.05h^{-1}$
X-ray halo of M87	> 750	$> 0.46h^{-1}$
Local Group timing	100	$0.06h^{-1}$
Groups of galaxies	$260h$	0.16
Clusters of galaxies	$400h$	0.25
Gravitational lenses	—	$0.1 - 0.3$
Big Bang nucleosynthesis	—	0.065 ± 0.045

The WMAP value of the density parameter for baryonic matter is

$$[\Omega_{b,0}]_{\text{WMAP}} = (0.0224 \pm 0.0009)h^{-2} = 0.044 \pm 0.004 \quad (\text{for } h = 0.71). \quad (29.22)$$

Thus, according to the WMAP results, baryonic matter accounts for only about 16% of the matter in the universe; the other 84% is some sort of nonbaryonic dark matter.

The general characteristics of the expansion of our model universe composed of pressureless dust can now be determined. First note that, from Eqs. (29.6) and (29.19),

$$\frac{\Omega}{\Omega_0} = \frac{\rho}{\rho_0} \frac{H_0^2}{H^2} = (1+z)^3 \frac{H_0^2}{H^2},$$

so

$$\Omega H^2 = (1+z)^3 \Omega_0 H_0^2. \quad (29.23)$$

Another relation between Ω and H comes from combining the density parameter, Eq. (29.18), with Eq. (29.9):

$$H^2(1 - \Omega)R^2 = -kc^2 \quad (29.24)$$

As a special case at $t = t_0$,

$$H_0^2(1 - \Omega_0) = -kc^2. \quad (29.25)$$

This confirms that:

- If $\Omega_0 > 1$, then $k > 0$ and the universe is closed.

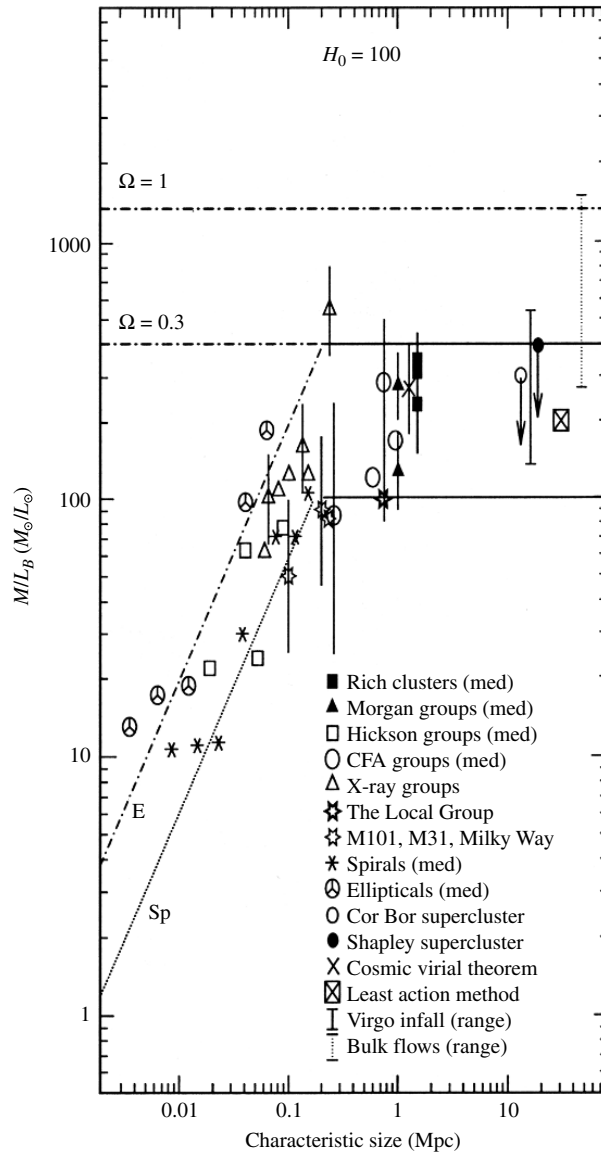


FIGURE 29.4 The mass-to-light ratio as a function of the characteristic size of a variety of systems. H_0 was taken to be $100 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for this figure prior to publication of the WMAP results. (Figure adapted from Dodelson, *Modern Cosmology*, Academic Press, New York, 2003, with permission from Elsevier. Data from Bahcall et al., *Ap. J.*, 541, 1, 2000.)

- If $\Omega_0 < 1$, then $k < 0$ and the universe is open.
- If $\Omega_0 = 1$, then $k = 0$ and the universe is flat.

Remember that we are now dealing with a simple model of a one-component universe of pressureless dust. Later we will study more realistic multicomponent models, which will show that a measurement of the mass density parameter alone is not enough for us to draw any conclusions about the ultimate fate of our physical universe.

Equating Eqs. (29.24) and (29.25), and using (29.4), we find

$$H^2(1 - \Omega) = H_0^2(1 - \Omega_0)(1 + z)^2. \quad (29.26)$$

Thus we have two equations, Eqs. (29.23) and (29.26), with the two unknowns Ω and H . These may be easily solved to find

$$H = H_0(1 + z)(1 + \Omega_0 z)^{1/2} \quad (29.27)$$

and

$$\Omega = \left(\frac{1 + z}{1 + \Omega_0 z} \right) \Omega_0 = 1 + \frac{\Omega_0 - 1}{1 + \Omega_0 z}. \quad (29.28)$$

Equation (29.27) implies that at very early times, as $R \rightarrow 0$ and $z \rightarrow \infty$, the Hubble parameter $H \rightarrow \infty$. Equation (29.28) shows that the sign of $\Omega - 1$ does not change, and in particular that if $\Omega = 1$ at any time, then $\Omega = 1$ at all times. The character of the universe does not change as the universe evolves; it is either always closed, always open, or always flat. Equation (29.28) also shows that at very early times, as $z \rightarrow \infty$, the density parameter $\Omega \rightarrow 1$ regardless of today's value of Ω_0 . *Therefore, the early universe was essentially flat*; see Fig. 29.5. The assumption of a flat early universe will greatly simplify the description of the first few minutes of the universe.

Example 29.1.1. Later we will find that when the universe was about 3 minutes old, protons and neutrons combined to form helium nuclei. This occurred at a redshift of $z = 3.68 \times 10^8$. Using the WMAP value of $[\Omega_{m,0}]_{\text{WMAP}} = 0.27$ for Ω_0 , we find that at the time of helium formation, the value of Ω was

$$\Omega = 1 + \frac{\Omega_0 - 1}{1 + \Omega_0 z} = 1 + \frac{0.27 - 1}{1 + (0.27)(3.68 \times 10^8)} = 0.99999999265. \quad (29.29)$$

At even earlier times the value of Ω contains a much longer string of nines. During the late twentieth century, it appeared absurd to theoreticians that a mechanism would exist to fine-tune Ω to a value so very close to unity without having an exactly flat universe with $\Omega = 1$. And yet, observational measurements of the value of the density parameter continued to hover around $\Omega_0 \simeq 0.3$. The solution to this puzzle will be described in Section 29.3, where we will find that the theoreticians and the observers were *both* correct.

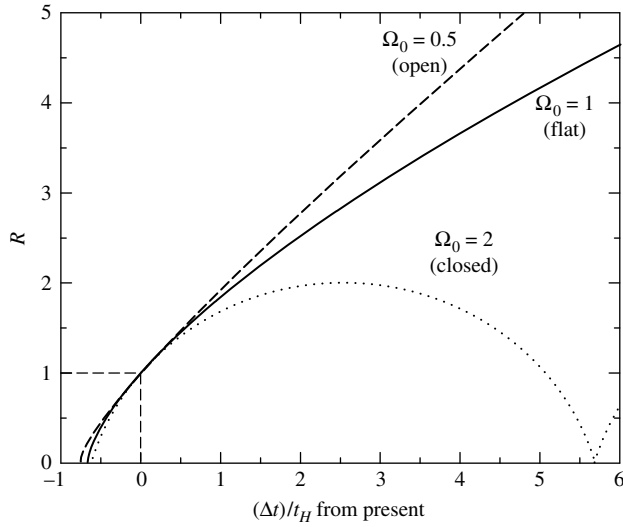


FIGURE 29.5 The evolution of the scale factor, R , for three model universes—open ($\Omega_0 = 0.5$), flat ($\Omega_0 = 1$), and closed ($\Omega_0 = 2$)—as a function of time, measured from the present. The dotted lines locate the position of today's universe on the three curves. At the present time ($R = 1$) all three universes have the same value of H_0 , as exhibited by the curves having the same slope. For the early universe ($R < 1$) there is little difference among the kinematic behaviors of a flat, a closed, and an open universe because the early universe was essentially flat. The elapsed time Δt is in units of the Hubble time, t_H .

The expansion of a flat, one-component universe of pressureless dust as a function of time may be found by solving Eq. (29.11) with $k = 0$ (so $\rho_0 = \rho_{c,0}$ and $\Omega_0 = 1$):

$$\left(\frac{dR}{dt}\right)^2 = \frac{8\pi G\rho_{c,0}}{3R}.$$

Taking the square root of each side and integrating (with $R = 0$ at $t = 0$) gives

$$\int_0^R \sqrt{R'} dR' = \sqrt{\frac{8\pi G\rho_{c,0}}{3}} \int_0^t dt'$$

or

$$R_{\text{flat}} = (6\pi G\rho_{c,0})^{1/3} t^{2/3} \quad (29.30)$$

$$= \left(\frac{3}{2}\right)^{2/3} \left(\frac{t}{t_H}\right)^{2/3} \quad (\text{for } \Omega_0 = 1), \quad (29.31)$$

where the last expression was obtained by using Eq. (29.15), and $t_H \equiv 1/H_0$ for the Hubble time (Eq. 27.14). The increase in R for $\Omega_0 = 1$ is shown in Fig. 29.5, with time in units of the Hubble time.

If $\Omega_0 \neq 1$, the density is not equal to the critical density and Eq. (29.11) is more difficult to solve. If $\Omega_0 > 1$, the universe is closed and the solution can be expressed in parametric form as

$$R_{\text{closed}} = \frac{4\pi G\rho_0}{3kc^2} [1 - \cos(x)] \quad (29.32)$$

$$= \frac{1}{2} \frac{\Omega_0}{\Omega_0 - 1} [1 - \cos(x)] \quad (29.33)$$

and

$$t_{\text{closed}} = \frac{4\pi G\rho_0}{3k^{3/2}c^3} [x - \sin(x)] \quad (29.34)$$

$$= \frac{1}{2H_0} \frac{\Omega_0}{(\Omega_0 - 1)^{3/2}} [x - \sin(x)], \quad (29.35)$$

where the variable $x \geq 0$ merely parameterizes the solution. [This may be easily verified by direct substitution into Eq. (29.11); see Problem 29.3.] The behavior of this solution with $\Omega_0 = 2$ is shown in Fig. 29.5. The “bounce” that occurs after the contraction of the universe is a mathematical artifact and does not imply an endless sequence of oscillating universes.

On the other hand, if $\Omega_0 < 1$, the universe is open and the parametric form of the solution of Eq. (29.11) is

$$R_{\text{open}} = \frac{4\pi G\rho_0}{3|k|c^2} [\cosh(x) - 1] \quad (29.36)$$

$$= \frac{1}{2} \frac{\Omega_0}{1 - \Omega_0} [\cosh(x) - 1] \quad (29.37)$$

and

$$t_{\text{open}} = \frac{4\pi G\rho_0}{3|k|^{3/2}c^3} [\sinh(x) - x] \quad (29.38)$$

$$= \frac{1}{2H_0} \frac{\Omega_0}{(1 - \Omega_0)^{3/2}} [\sinh(x) - x]. \quad (29.39)$$

Recall that the hyperbolic cosine is defined as $\cosh(x) \equiv (e^x + e^{-x})/2 \geq 1$. Similarly, the hyperbolic sine is given by $\sinh(x) \equiv (e^x - e^{-x})/2 \geq x$, so R_{open} increases monotonically with t . See Fig. 29.5 for the appearance of the solution with $\Omega_0 = 0.5$. If $\Omega_0 \leq 1$, then the universe will continue to expand forever.

The Age of the Pressureless “Dust” Universe

We are now ready to calculate the age of the universe as a function of the redshift z . Before continuing, a note of caution should be sounded about referring to any time t as the “age of the universe.” The laws of physics, as we presently understand them, cannot remain valid under the extreme conditions that must prevail as $t \rightarrow 0$. In using t as a measure of the time since the Big Bang, we must always keep in mind that this is an *extrapolated time* and cannot be taken literally at the earliest instants ($t < 10^{-43}$ s; see Section 30.1).

Keeping this admonition in mind, we now proceed by using Eq. (29.4) to replace R by $1/(1+z)$ in Eq. (29.31) for a flat universe. The age of a flat universe (in units of the Hubble time) that is observed at redshift z is then found to be

$$\frac{t_{\text{flat}}(z)}{t_H} = \frac{2}{3} \frac{1}{(1+z)^{3/2}} \quad (\text{for } \Omega_0 = 1). \quad (29.40)$$

Replacing R by $1/(1+z)$ in Eq. (29.33) for a closed universe and using Eq. (29.35) to eliminate x leads to

$$\frac{t_{\text{closed}}(z)}{t_H} = \frac{\Omega_0}{2(\Omega_0 - 1)^{3/2}} \left[\cos^{-1} \left(\frac{\Omega_0 z - \Omega_0 + 2}{\Omega_0 z + \Omega_0} \right) - \frac{2\sqrt{(\Omega_0 - 1)(\Omega_0 z + 1)}}{\Omega_0(1+z)} \right] \quad (\text{for } \Omega_0 > 1). \quad (29.41)$$

Following a similar procedure using Eq. (29.37) for an open universe and using Eq. (29.39) to eliminate x results in

$$\frac{t_{\text{open}}(z)}{t_H} = \frac{\Omega_0}{2(1 - \Omega_0)^{3/2}} \left[-\cosh^{-1} \left(\frac{\Omega_0 z - \Omega_0 + 2}{\Omega_0 z + \Omega_0} \right) + \frac{2\sqrt{(1 - \Omega_0)(\Omega_0 z + 1)}}{\Omega_0(1+z)} \right] \quad (\text{for } \Omega_0 < 1). \quad (29.42)$$

In the limit of large redshift, Eqs. (29.40) through (29.42) all reduce to

$$\frac{t(z)}{t_H} = \frac{2}{3} \frac{1}{(1+z)^{3/2} \Omega_0^{1/2}}, \quad (29.43)$$

where the higher-order terms may be neglected for $\Omega_0 \neq 1$ (see Problem 29.11). Because the early universe was flat to a very good approximation, precise observations are required to determine whether the universe is flat, closed, or open. In Section 29.4, the observational aspects of cosmology will be considered in more detail.

The current age of the universe, t_0 , may be easily found by setting $z = 0$ in Eqs. (29.40–29.42) to find

$$\frac{t_{\text{flat},0}}{t_H} = \frac{2}{3} \quad (\text{for } \Omega_0 = 1) \quad (29.44)$$

for a flat universe,

$$\frac{t_{\text{closed},0}}{t_H} = \frac{\Omega_0}{2(\Omega_0 - 1)^{3/2}} \left[\cos^{-1} \left(\frac{2}{\Omega_0} - 1 \right) - \frac{2\sqrt{\Omega_0 - 1}}{\Omega_0} \right] \quad (\text{for } \Omega_0 > 1) \quad (29.45)$$

for a closed universe, and

$$\frac{t_{\text{open},0}}{t_H} = \frac{\Omega_0}{2(1 - \Omega_0)^{3/2}} \left[-\cosh^{-1} \left(\frac{2}{\Omega_0} - 1 \right) + \frac{2\sqrt{1 - \Omega_0}}{\Omega_0} \right] \quad (\text{for } \Omega_0 < 1) \quad (29.46)$$

for an open universe.

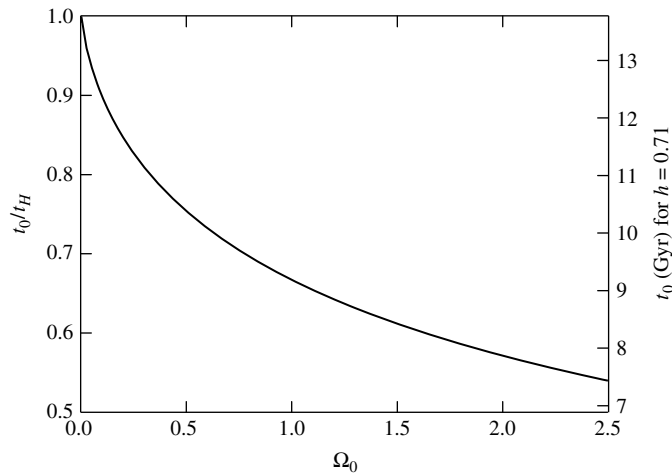


FIGURE 29.6 The age of the universe as a function of the density parameter, Ω_0 . The age is expressed as a fraction of the Hubble time, $t_H \simeq 10^{10}h^{-1}$ yr. The right axis shows the age in billions of years for $h = 0.71$.

The age of the universe for these models, expressed as a fraction of the Hubble time, is shown in Fig. 29.6. As we will see in Section 30.1, according to the inflation scenario the universe should be essentially flat, a scenario supported by recent observations. If the average density of the universe is equal to the critical density, then the age of the universe is two-thirds of the Hubble time. Using the WMAP value of $[h]_{\text{WMAP}} = 0.71$ gives an age of about 9.2 Gyr. Although this result is less than the currently accepted value of 13.7 Gyr, it is remarkable that this simple model of an expanding universe of pressureless dust produces ages that are in rough accordance with the mean age of the oldest globular clusters, 11.5 billion years.⁷

Piecing together the history of the universe is a bit like working a jigsaw puzzle. As progress is made and the pattern begins to emerge, the shapes of the missing pieces are defined by those already in place. For our simple model of a pressureless dust universe, the pieces overlap a bit; the age of the oldest stars is greater than the age of the universe. However, there is more to the universe than pressureless dust. For example, the universe is filled with photons—about two billion photons for every baryon. Photons, baryons, and other constituents will play a role in resolving the discrepancy between the age of the oldest globular clusters and the age of the universe.

The Lookback Time

The **lookback time**, t_L , is defined as how far back in time we are looking when we view an object with redshift z . This is just the difference between the present age of the universe and its age at time $t(z)$,

$$t_L = t_0 - t(z). \quad (29.47)$$

⁷A more accurate comparison between the age of the universe and the ages of globular clusters will be made in Section 29.3.

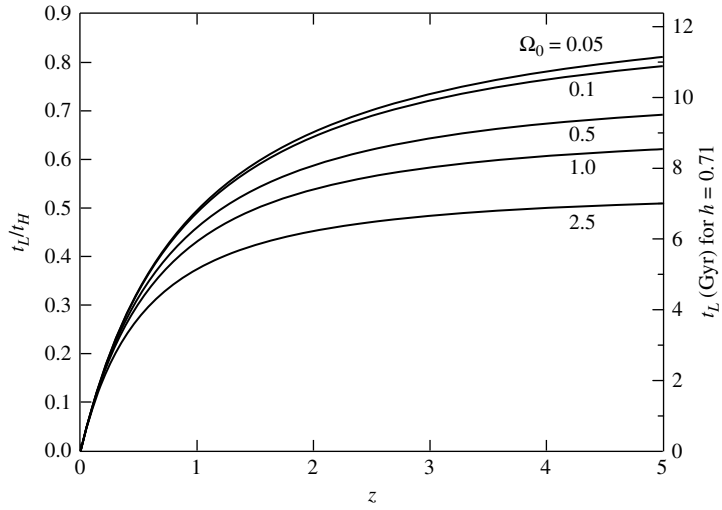


FIGURE 29.7 The lookback time as a function of the redshift, z , for a range of values of the density parameter, Ω_0 . The lookback time is expressed as a fraction of the Hubble time, $t_H \simeq 10^{10} h^{-1}$ yr. The right axis shows the lookback time in billions of years if $h = 0.71$.

For example, for a flat universe, Eqs. (29.40) and (29.44) show that the lookback time is, in units of the Hubble time,

$$\frac{t_L}{t_H} = \frac{2}{3} \left[1 - \frac{1}{(1+z)^{3/2}} \right] \quad (\text{for } \Omega_0 = 1). \quad (29.48)$$

Figure 29.7 shows the lookback times for flat, closed, and open models of the universe.

Example 29.1.2. In Example 4.3.2, the redshift of the quasar SDSS 1030+0524 was found to be $z = 6.28$. Assuming a flat universe of pressureless dust, Eq. (29.48) shows that the lookback time to this quasar is

$$\frac{t_L}{t_H} = \frac{2}{3} (1 - 0.0509) = 0.633.$$

Since the age of a flat universe is $t_0 = 2t_H/3$,

$$\frac{t_L}{t_0} = 0.949.$$

This means that only 5% of the history of the universe had unfolded when the light left this quasar. At that time the universe was smaller by about a factor of 7 when, according to Eq. (29.4), the scale factor was

$$R = \frac{1}{1+z} = 0.137.$$

Extending Our Simple Model to Include Pressure

Let's take stock of the basic equations we have derived so far and generalize them a bit, anticipating some features of the equations of general relativity we will encounter in Section 29.3. We start with Eq. (29.10),

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3} \pi G \rho \right] R^2 = -kc^2.$$

With a nod toward Einstein's $E_{\text{rest}} = mc^2$, we broaden the meaning of the density ρ to include matter in all of its forms. For nonrelativistic particles, ρ is the usual mass density. For relativistic particles, such as photons and neutrinos, ρ is the **equivalent mass density**—that is, the energy density divided by c^2 .

Equation (29.5), $R^3 \rho = \rho_0$, describes the conservation of mass within the expanding shell. Again acknowledging the equivalence of mass and energy, this equation is also a statement of the conservation of energy for a pressureless dust universe.

A thermodynamic argument supplies the generalization of Eq. (29.5) for models of the universe that incorporate pressure-producing components. Imagine a universe filled with a fluid (dust, photons, etc.) of uniform density ρ , pressure P , and temperature T , and choose an arbitrary point for the origin. Let r be the radius of a comoving spherical surface, centered on the origin.⁸ We will employ the first law of thermodynamics, which applies the law of conservation of internal energy, U , to the fluid within the expanding sphere:

$$dU = dQ - dW. \quad (29.49)$$

First note that the entire universe has the same temperature, so there can be no heat flow: $dQ = 0$. That is, the expansion of the universe is adiabatic. Any change in internal energy must be produced by work done by the fluid. Writing the result as a time derivative,

$$\frac{dU}{dt} = -\frac{dW}{dt} = -P \frac{dV}{dt}.$$

and substituting $V = \frac{4}{3}\pi r^3$, we obtain

$$\frac{dU}{dt} = -\frac{4}{3}\pi P \frac{d(r^3)}{dt}.$$

If we define the internal energy per unit volume u as

$$u = \frac{U}{\frac{4}{3}\pi r^3},$$

then we find

$$\frac{d(r^3 u)}{dt} = -P \frac{d(r^3)}{dt}.$$

⁸We need not worry about photons leaving the sphere, because the cosmological principle assures us that the same number of equivalent photons will enter the sphere.

Writing u in terms of the equivalent mass density ρ ,

$$\rho = \frac{u}{c^2},$$

gives

$$\frac{d(r^3 \rho)}{dt} = -\frac{P}{c^2} \frac{d(r^3)}{dt}.$$

Finally, using $r = R\varpi$ (Eq. 29.3), we obtain the **fluid equation**,

$$\frac{d(R^3 \rho)}{dt} = -\frac{P}{c^2} \frac{d(R^3)}{dt}. \quad (29.50)$$

For a universe of pressureless dust, $P = 0$ so $R^3 \rho = \text{constant}$, in agreement with Eq. (29.5).

An equation describing the acceleration of the universal expansion can be obtained by multiplying Eq. (29.10) by R and then taking a time derivative. Using Eq. (29.50) to replace $d(\rho R^3)/dt$ and using Eq. (29.10) to eliminate the $-kc^2$, we arrive at the **acceleration equation**

$$\frac{d^2 R}{dt^2} = -\frac{4}{3}\pi G \left(\rho + \frac{3P}{c^2} \right) R. \quad (29.51)$$

Note that the effect of the pressure P is to slow down the expansion (assuming $P > 0$). If this seems counterintuitive, recall that because the pressure is the same everywhere in the universe, both inside and outside the shell, there is no pressure gradient to exert a net force on the expanding sphere. The answer lies in the motion of the particles that creates the fluid's pressure. The equivalent mass of the particles' kinetic energy creates a gravitational attraction that slows down the expansion just as their actual mass does. In fact, the assumption that $P = 0$ is valid for much of the history of the universe. For instance, you will find in Problem 29.14 that $\rho \gg P/c^2$ in today's universe.

Equation (29.51) is an illustration of **Birkhoff's theorem**. In 1923 the American mathematician G. D. Birkhoff (1884–1944) proved quite generally that for a spherically symmetric distribution of matter, Einstein's field equations have a unique solution. As a corollary, the acceleration of an expanding shell in our fluid universe is determined solely by the fluid lying within the shell.⁹ Equation (29.51) shows that the acceleration does not depend on any factors other than ρ , P , and R . Because Birkhoff's theorem holds even when general relativity is included, it is quite important in the study of cosmology.

Equations (29.10), (29.50), and (29.51) have three unknowns: R , ρ , and P . However, the equations are not independent; any two may be used to derive the third. To solve these two equations for R , ρ , and P , we need a third relation, an *equation of state*, that links the variables. Such an equation of state can be written generally as

$$P = wu = w\rho c^2, \quad (29.52)$$

⁹You may recall that a similar result was obtained in Problem 10.2.

where w is a constant. In words, the pressure is proportional to the energy density of the fluid. For example, for mass in the form of pressureless dust, $w_m = 0$, and for blackbody radiation, with the equation of state $P_{\text{rad}} = u_{\text{rad}}/3$ (Eq. 9.11), we have $w_{\text{rad}} = 1/3$. Inserting the general equation of state, Eq. (29.52), into the fluid equation, Eq. (29.50), quickly produces the relation (Problem 29.15)

$$R^{3(1+w)} \rho = \text{constant} = \rho_0, \quad (29.53)$$

where ρ_0 is the present value of the mass density (or equivalent mass density). For pressureless dust ($w_m = 0$), we recover Eq. (29.5), $R^3 \rho_m = \rho_{m,0}$.

The Deceleration Parameter

Finally, we introduce a useful dimensionless quantity that describes the acceleration of the universal expansion: the **deceleration parameter**, $q(t)$, which is defined as

$$q(t) \equiv -\frac{R(t) [d^2 R(t)/dt^2]}{[dR(t)/dt]^2}. \quad (29.54)$$

Both the name and the minus sign (to ensure that $q > 0$ for a deceleration) betray the certainty of twentieth-century astronomers that the expansion of the universe must be slowing down with time.¹⁰ It is left to Problem 29.16 to show that for a pressureless dust universe,

$$q(t) = \frac{1}{2} \Omega(t), \quad (29.55)$$

and so at the present time,

$$q_0 = \frac{1}{2} \Omega_0. \quad (29.56)$$

Thus, for a pressureless dust universe, $q_0 = 0.5$ for a flat universe, while $q_0 > 0.5$ and $q_0 < 0.5$ correspond to a closed and an open universe, respectively.

29.2 ■ THE COSMIC MICROWAVE BACKGROUND

In 1946 George Gamow was pondering the cosmic abundances of the elements. Realizing that the newborn, dense universe must have been hot enough for a burst of nuclear reactions to occur, he proposed that a sequence of reactions in the very early universe could explain the measured cosmic abundance curve (shown in Fig. 15.16). Gamow, together with Ralph Alpher, published this idea two years later.¹¹ Still later, however, detailed calculations by

¹⁰In Section 29.3 this sense of certainty will be toppled.

¹¹Gamow, a Russian émigré, was famous for his Puckish sense of humor. When this paper first appeared, Gamow added Hans Bethe as a co-author (without Bethe's knowledge). Gamow thought it would be appropriate that a paper on cosmic beginnings be authored by Alpher (α), Bethe (β), and Gamow (γ), a play on the first three letters of the Greek alphabet.

Alpher and Robert Herman (1914–1997) showed that Gamow’s idea was flawed because there were roadblocks to assembling successively heavier nuclei simply by adding protons or neutrons. There are no stable nuclei with five or eight nucleons, leaving ${}^4_2\text{He}$ as the heaviest element that can be formed as Gamow proposed. (A small amount of an isotope of lithium, ${}^7_3\text{Li}$, is also formed in the early universe by the nuclear fusion of ${}^4_2\text{He}$ with ${}^3_1\text{H}$ and the fusion of ${}^4_2\text{He}$ with ${}^3_2\text{He}$. The latter produces ${}^7_4\text{Be}$, which radioactively decays to ${}^7_3\text{Li}$.)

At that time, there was also a large problem with the idea of a hot, dense universe coming into existence approximately one Hubble time ago. Edwin Hubble’s original value of his constant was $H_0 = 500 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (see Fig. 27.7), which corresponds to $t_H = 1/H_0 = 10^9 \text{ yr}$ for the age of the universe. This is only a fraction of Earth’s age, which in 1928 had been radioactively dated as several billion years. By the late 1940s further measurements indicated that $1/H_0 = 1.8 \times 10^9 \text{ yr}$, which was still embarrassingly short. It was certainly hard to understand how Earth could be older than the universe.

The Steady-State Model of the Universe

In 1946 at Cambridge University, Hermann Bondi (1919–2005), Thomas Gold (1920–2004), and Fred Hoyle (1915–2001) attempted to find an alternative to Gamow’s unpalatable Big Bang universe.¹² In papers published in 1948 and 1949, they proposed their model of a **steady-state universe**. It extended the cosmological principle to include time, stating that, in addition to the universe being isotropic and homogeneous, it also appears the same *at all times*. A steady-state universe has no beginning and no end. It is infinitely old, and as it expands, a continuous creation of matter is required to maintain the average density of the universe at its present level. This changes the interpretation of the Hubble time; rather than the characteristic age of the universe, t_H becomes a characteristic time for the creation of matter. If the universe roughly doubles in size in time t_H , then its volume becomes eight times greater, and so the rate of matter creation required to maintain the universe as it is today is approximately $8\rho_0/t_H = 8H_0\rho_0$. Just a few hydrogen atoms per cubic meter of space would need to be created every ten billion years, a rate far too small to be measured experimentally. In the original steady-state models, the “when,” “where,” and “how” of the spontaneous appearance of new matter (in violation of the law of conservation of mass–energy) were questions left unanswered. The appeal of the steady-state universe was its resolution of the timescale problem.¹³

Just as Gamow and his collaborators tried to explain the cosmic abundance curve (Fig. 15.16) by using the nuclear reactions that would accompany a Big Bang, Hoyle sought an explanation in the nuclear reactions that took place inside stars. He joined forces with two English colleagues, Geoffrey and Margaret Burbidge (a theoretical physicist and an astronomer, respectively), and William Fowler (1911–1995), an American physicist. In

¹²Ironically, it was Fred Hoyle who came up with the term *Big Bang*. He used it derisively in a 1950 BBC radio broadcast when he said, “This big bang idea seemed to me to be unsatisfactory even before examination showed that it leads to serious difficulties. For when we look at our own Galaxy there is not the smallest sign that such an explosion ever occurred.”

¹³The solution to the short Hubble time appeared in 1952, when Walter Baade discovered that there were two types of Cepheid variable stars. This revised the period–luminosity relation that forms the foundation of the cosmic distance scale; see Section 27.1.

1957 they published their seminal paper, referred to as B²FH, that laid out the theory of stellar nucleosynthesis as described in Chapters 10 and 13.

The B²FH analysis was a success, and its results were compatible with both the Big Bang and steady-state cosmologies. During the 1950s both theories had their supporters and detractors. However, in addition to the unanswered questions about the continuous creation of matter, the steady-state theory had a serious problem explaining the large amount of helium observed in the universe. Astronomers had established that about one-quarter (0.274 ± 0.016) of the baryonic mass of the universe is in the form of helium. When compared to the cosmic abundances of the heavier elements, it was clear that stellar nucleosynthesis could not account for the amount of helium observed, especially considering that carbon, nitrogen, and oxygen are the results of exhaustively burning the star's helium core (recall Section 13.2). Gamow, Alpher, and Herman had shown that the Big Bang could at least explain the abundance of helium, but where was the proof that such a violent event had ever occurred? To invoke an event that was apparently beyond the reach of investigation seemed unscientific to many astronomers.

The Cooling of the Universe after the Big Bang

A key idea of the α - β - γ paper was that the dense, early universe must have been very hot. In this hot, dense universe, the mean free path of photons would have been short enough to maintain thermodynamic equilibrium. Although an expanding universe cannot be precisely in equilibrium, this assumption of thermodynamic equilibrium is extremely good. As was described in Section 9.2, under these conditions the radiation field has a blackbody spectrum. In 1948 Alpher and Herman published their description of how this blackbody radiation would have cooled as the universe expanded, and they predicted that the universe should now be filled with blackbody radiation at a temperature of 5 K.

The cooling of the blackbody radiation can be derived by considering its energy density $u = aT^4$ (Eq. 9.7). According to the fluid equation (Eq. 29.53) with $w_{\text{rad}} = 1/3$ for blackbody radiation and $R(t_0) = 1$,

$$R^{3(1+w_{\text{rad}})} u_{\text{rad}} = R^4 u_{\text{rad}} = u_{\text{rad},0}. \quad (29.57)$$

The energy density today, $u_{\text{rad},0}$, is smaller than the earlier value u_{rad} by a factor of R^4 ; a factor of R^3 is due to the fact that the volume of the universe has increased since then, and the other factor of R comes from the lesser energy of today's longer-wavelength photons ($E_{\text{photon}} = hc/\lambda$), a result of the cosmological redshift. Thus

$$R^4 a T^4 = a T_0^4,$$

and we find that the present blackbody temperature must be related to the temperature at an earlier time by

$$RT = T_0. \quad (29.58)$$

That is, the product of the scale factor and the blackbody temperature remains constant as the universe expands. When the universe was half as large, it was twice as hot.

An order-of-magnitude estimate of the present blackbody temperature of the universe may be calculated by considering the temperature and baryonic mass density that must have prevailed in the early universe when helium was being formed. The fusion of hydrogen nuclei requires roughly that $T \simeq 10^9$ K and $\rho_b \simeq 10^{-2}$ kg m⁻³. If the temperature were any higher, the deuterium nuclei involved in the fusion chain would have undergone photodissociation due to the presence of energetic blackbody radiation, whereas a lower temperature would have made the Coulomb barrier between the nuclei too difficult to overcome. The quoted density is needed to produce the observed amounts of ${}^3\text{He}$ and other nuclei. From Eqs. (29.5) and (29.17), the value of the scale factor at the time of the helium formation was roughly

$$R \simeq \left(\frac{\rho_{b,0}}{\rho_b} \right)^{1/3} = 3.47 \times 10^{-9}.$$

At that time, the universe was only a few billionths of its present size. Combining the scale factor with $T(R) = 10^9$ K, the present temperature of the blackbody radiation can be estimated from Eq. (29.58) as

$$T_0 = RT(R) \simeq 3.47 \text{ K},$$

similar to Alpher and Hermann's original estimate of 5 K in 1948. Wien's law then gives the peak wavelength of the blackbody spectrum as

$$\lambda_{\text{max}} = \frac{0.00290 \text{ m K}}{T_0} \simeq 8.36 \times 10^{-4} \text{ m}.$$

The Discovery of the Cosmic Microwave Background

Sixteen years after Alpher and Herman predicted that the universe had cooled to 5 K and was filled with blackbody radiation, Robert Dicke (1916–1997) of Princeton University and his postdoctoral student P. J. E. Peebles unknowingly followed in their footsteps. In 1964 Peebles calculated that the blackbody radiation left over from the Big Bang should have a temperature of about 10 K. Unlike Alpher and Herman, however, Dicke was interested in searching for the relic radiation. He didn't realize that this cosmic background radiation had just been found by two radio astronomers working a few miles away at Bell Laboratories in Holmdel, New Jersey.¹⁴ Arno Penzias and Robert Wilson were working with a huge horn reflector antenna (shown in Fig. 29.8) that had been used to communicate with the new Telstar satellite. Despite a year of effort, the two men had been unable to get rid of a persistent hiss in the signal. The hiss came continually from all directions in the sky and remained even after Penzias and Wilson had scrubbed their antenna clean, taped over seams and rivets, and removed two pigeons that had nested inside the horn.¹⁵ They knew that a 3 K blackbody would produce their interference but were unaware of any possible source until Penzias learned of Peeble's calculation of a 10-K background. Penzias called

¹⁴Interestingly, it was in Holmdel that Karl Jansky built his first radio telescope.

¹⁵The pigeons were freed 60 miles away. Unfortunately, they proved to be homing pigeons and had to be removed again.

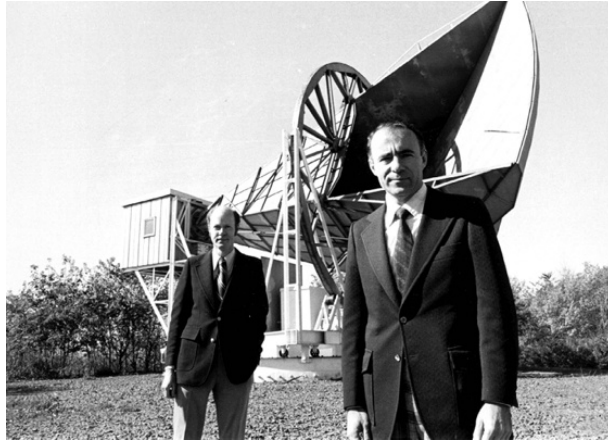


FIGURE 29.8 Robert Wilson and Arno Penzias standing in front of the antenna used to first identify the cosmic microwave background. (Courtesy of AT&T Archives.)

Dicke and invited him to Holmdel, and in 1965 the pieces of the puzzle finally fell together. Penzias and Wilson had detected the blackbody radiation that fills the universe, with a peak wavelength of $\lambda_{\text{max}} = 1.06 \text{ mm}$ in the microwave region of the electromagnetic spectrum. This afterglow of the Big Bang is now known as the **cosmic microwave background**, often abbreviated as the CMB. Dicke, Peebles, and their co-workers at Princeton immediately wrote a note for the *Astrophysical Journal Letters* detailing the theory of the cosmic microwave background that strongly supported the Big Bang cosmology, while Penzias and Wilson wrote an accompanying letter, discreetly titled “A Measurement of Excess Antenna Temperature at 4080 Megacycles per Second,” that described their discovery.

The discovery of the CMB was a death knell for steady-state cosmology. As further measurements at other wavelengths confirmed that the shape of the CMB spectrum was that of a blackbody, the number of astronomers supporting the Big Bang theory swelled while those favoring a steady-state universe dwindled. In 1991 a striking measurement of the cosmic microwave background was obtained by the COBE satellite (see Section 6.4). The COBE measurement of the spectrum of the CMB is shown in Fig. 29.9. The data points (whose errors are smaller than the points themselves) fall almost perfectly on the theoretical spectrum of a 2.725-K blackbody. The Planck function $B_\nu(T)$ in Fig. 29.9 peaks at a frequency of 160 GHz, corresponding to the frequency version of Wien’s law,¹⁶

$$\frac{\nu_{\text{max}}}{T} = 5.88 \times 10^{10} \text{ Hz K}^{-1}. \quad (29.59)$$

The WMAP value for the CMB is

$$[T_0]_{\text{WMAP}} = 2.725 \pm 0.002 \text{ K}, \quad (29.60)$$

remarkably close to our estimate of 3.47 K obtained from the simple estimate on page 1165.

¹⁶This version of Wien’s law may be found from Eq. (3.24) by setting $dB_\nu(T)/d\nu = 0$.

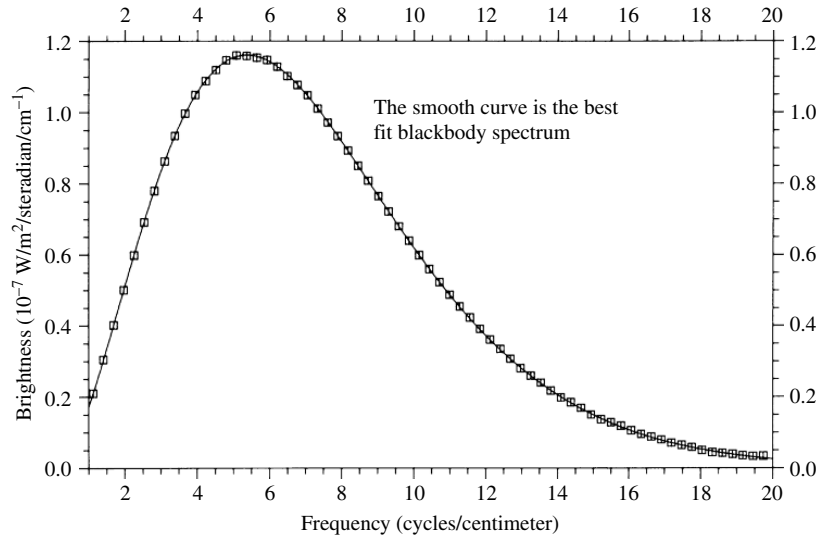


FIGURE 29.9 The COBE measurement of the spectrum of the cosmic microwave background, which is that of a blackbody with a temperature of 2.725 K. The horizontal axis (frequency) is actually $1/\lambda$ (cm^{-1}); the spectrum peaks at a frequency of 160 GHz (5.35 cycles per centimeter). (Figure adapted from Mather et al., *Ap. J. Lett.*, 354, L37, 1990. Courtesy of NASA/GSFC and the COBE Science Working Group.)

The Dipole Anisotropy of the CMB

The cosmic microwave background suffuses the entire universe. It does not emanate from any object but, rather, originated in the Big Bang, when the entire universe was essentially located at a single point (or event) of spacetime. For this reason, all observers at rest with respect to the Hubble flow (no peculiar velocity) see the same spectrum for the CMB, with the same intensity in all directions (the CMB is isotropic). In particular, two observers in different galaxies that are being carried apart by the Hubble flow see the same blackbody spectrum.¹⁷

However, there is a Doppler shift of the CMB caused by an observer's peculiar velocity through space, relative to the Hubble flow. Using Wien's law, a shift in wavelength can be expressed as a change in the temperature of the blackbody radiation. For example, a slight blueshift (smaller λ_{max}) would correspond to a slightly higher temperature. Suppose an observer at rest relative to the Hubble flow determines that the cosmic microwave background has a temperature T_{rest} . Then, as shown in Problem 29.21, the temperature measured by an observer with a peculiar velocity v relative to the Hubble flow is

$$T_{\text{moving}} = \frac{T_{\text{rest}} \sqrt{1 - v^2/c^2}}{1 - (v/c) \cos \theta}, \quad (29.61)$$

¹⁷It is assumed that the two observers agree on the age of the universe when their measurements are made. Of course, each galaxy views the other as it appeared at an earlier time.

where θ is the angle between the direction of observation and the direction of motion. Both observers see a blackbody spectrum, but the moving observer measures a slightly hotter temperature in the forward direction ($\theta = 0$) and a slightly cooler temperature in the opposite direction. If the peculiar velocity is $v \ll c$, then

$$T_{\text{moving}} \simeq T_{\text{rest}} \left(1 + \frac{v}{c} \cos \theta \right) \quad (29.62)$$

(the proof is left as an exercise). The second term on the right-hand side, called the **dipole anisotropy** of the CMB, has been detected and measured; see Fig. 29.10. The temperature variation indicates that the peculiar velocity of the Sun relative to the Hubble flow is $370.6 \pm 0.4 \text{ km s}^{-1}$ in the direction $(\alpha, \delta) = (11.2^{\text{h}}, -7^{\circ})$, between the constellations of Leo and Crater. Of course, the Sun is orbiting the Galaxy, and the Milky Way is moving within the Local Group of galaxies. When these motions are accounted for, the peculiar motion of the Local Group relative to the Hubble flow is about 627 km s^{-1} toward $(\alpha, \delta) = (11.1^{\text{h}}, -27^{\circ})$, in the middle of the constellation Hydra. From this observation and measurements of the velocities of other galaxies and clusters of galaxies, astronomers have discovered a large-scale streaming motion of thousands of galaxies at $\sim 600 \text{ km s}^{-1}$ in the direction of $(\alpha, \delta) = (13.3^{\text{h}}, -44^{\circ})$ in the constellation Centaurus. The Hydra–Centaurus supercluster is also being carried along in this riverlike perturbation of the Hubble flow; see Section 27.3.

After the dipole anisotropy has been subtracted from the CMB, the remaining radiation is incredibly isotropic, having nearly equal intensity in all directions. Sensitive instruments, however, have revealed that the CMB does have hotter and cooler areas. The CMB appears as a patchwork of small regions, about 1° degree or less in diameter, where the temperature departs from the average value (T_0) by about one part in 10^5 . Careful observations and analyses of these regions by WMAP and various ground-based and balloon-borne experiments have produced the first precision measurement of cosmological parameters, as will

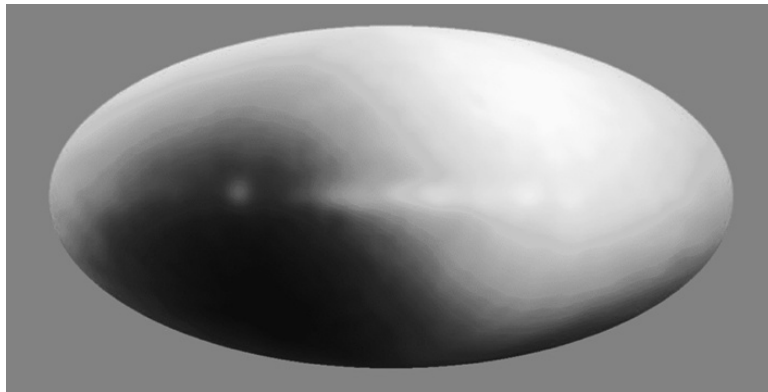


FIGURE 29.10 The dipole anisotropy in the CMB caused by the Sun's peculiar velocity, shown in Galactic coordinates. The figure is a combination of observations made at 53 GHz and 90 GHz. Lighter shades are somewhat hotter, and darker shades are somewhat cooler, than the 2.725-K CMB. The horizontal bright feature is due to Galactic sources. (Figure from Smoot et al., *Ap. J. Lett.*, 371, L1, 1991.)

be described in Section 30.2. We will continue to use the WMAP values of these parameters throughout this and the next chapter.

The Sunyaev–Zel’dovich Effect

It should be emphasized that an observer in a galaxy being carried along with the Hubble flow (no peculiar velocity) does not measure a Doppler shift of the CMB. An observer in a distant galaxy receding from us at an appreciable fraction of the speed of light sees the same CMB spectrum that we do. Evidence of this is produced when low-energy photons of the CMB pass through the hot ($\simeq 10^8$ K) ionized intracluster gas in a rich cluster of galaxies. A small fraction (typically 10^{-3} to 10^{-2}) of the photons are scattered to higher energies by the high-energy electrons in the gas. This **inverse Compton scattering** increases the frequency of a scattered photon by an average amount $\Delta\nu$ of

$$\frac{\overline{\Delta\nu}}{\nu} = 4 \frac{kT_e}{m_e c^2}, \quad (29.63)$$

where T_e is the temperature of the electron gas (see Problem 29.23). The resulting distortion of the CMB spectrum, shown in Fig. 29.11, is called the thermal **Sunyaev–Zel’dovich effect** [named for two Russians, astrophysicist Rashid Sunyaev and physicist Yakov Zel’dovich (1914–1987)].¹⁸ Although the spectrum no longer has the precise shape of a blackbody, its translation to higher frequencies may be used to define an effective decrease ΔT in the temperature T_0 of the CMB of approximately¹⁹

$$\frac{\Delta T}{T_0} \simeq -2 \frac{kT_e}{m_e c^2} \tau \quad (29.64)$$

where τ is the optical depth of the intracluster gas along the line of sight. Typical values of $\Delta T/T_0$ are a few times 10^{-4} . Observations of the Sunyaev–Zel’dovich effect for many clusters of galaxies confirm that it is independent of the cluster’s redshift, as expected if the CMB spectrum observed at a cluster is not affected by the cluster’s recessional velocity. Figure 29.12 shows the Sunyaev–Zel’dovich effect surrounding two clusters of galaxies. In addition to confirming the cosmological nature of the CMB, the Sunyaev–Zel’dovich effect is a promising probe of the properties and evolution of rich clusters of galaxies in the early universe.

Does the CMB Constitute a Preferred Frame of Reference?

It may seem to you that the frame of reference in which the CMB appears isotropic is a preferred frame of reference that is *truly motionless*, in violation of the postulates of special relativity as described in Section 4.2. We should remember, however, that general relativity is concerned with *local* inertial reference frames (cf. Section 17.1). There is no single frame

¹⁸If the cluster has a peculiar velocity, the bulk motion of the intracluster gas produces an additional Doppler shift of the scattered photons. This smaller perturbation of the CMB spectrum is called the kinetic Sunyaev–Zel’dovich effect.

¹⁹The leading coefficient in Eq. (29.64) is equal to -2 only on the Rayleigh–Jeans (long-wavelength) side of the spectrum; cf. Eq. (3.20).

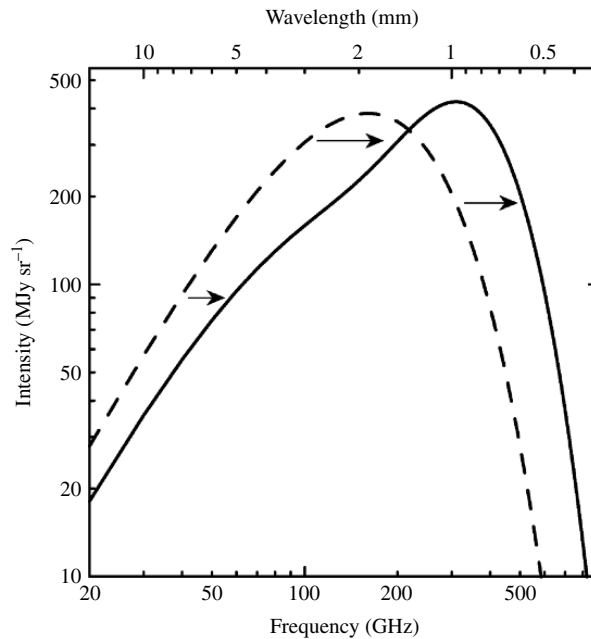


FIGURE 29.11 The undistorted CMB spectrum (dashed line) and the spectrum distorted by the Sunyaev–Zel’dovich effect (solid line). In a rich cluster of galaxies, CMB photons may be scattered to higher frequencies by colliding with the electrons in the hot intracluster gas. For frequencies less than the peak frequency, more photons are scattered out of a frequency interval than into it, so the intensity at that frequency decreases. Similarly, for frequencies greater than the peak frequency, fewer photons are scattered out of a frequency interval than into it, so the intensity at that frequency increases. The net result is a shift of the CMB spectrum to higher frequencies. The calculated distortion has been exaggerated by employing a fictional cluster 1000 times more massive than a typical rich cluster of galaxies. (Figure adapted from Carlstrom, Holder, and Reese, *Annu. Rev. Astron. Astrophys.*, 40, 646, 2002. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 40, ©2002 by Annual Reviews Inc.)

of reference that is capable of covering the entire universe. Although we can define a local reference frame at rest relative to the glow of the CMB, we can also define a local reference frame at rest relative to the motions of nearby stars (the local standard of rest described in Section 24.3). Both of these local inertial reference frames measure velocities relative to an arbitrary standard of rest and do not violate the tenets of relativity.

A Two-Component Model of the Universe

Recall that in Section 29.1 we considered the expansion of a universe with a single component, pressureless dust, that was slowing down due to its own self-gravity. According to the relativistic equivalence of mass and energy, however, the effect of the cosmic microwave background on the expansion must also be included. In fact, we now know that the gravitational effect of the CMB photons dominated the dynamics of the early universe, although their effect is completely negligible in the present universe.

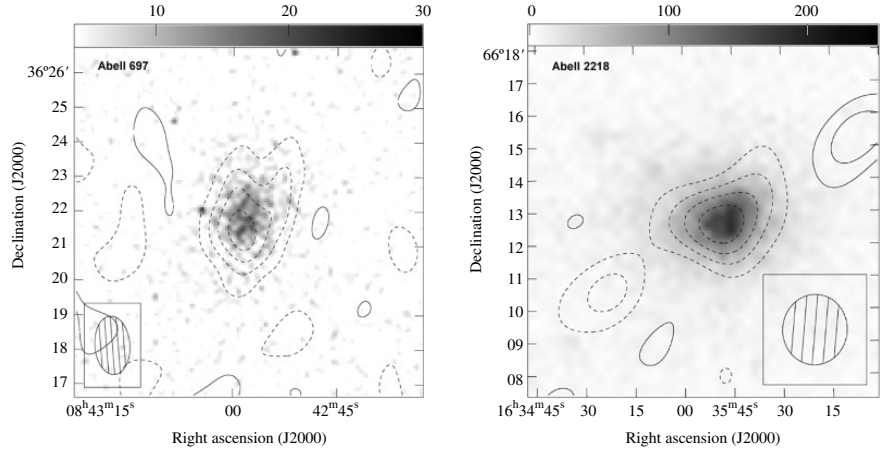


FIGURE 29.12 Radio contours showing the Sunyaev-Zel'dovich effect superimposed on ROSAT images of the clusters Abell 697 ($\Delta T = 1047 \mu\text{K}$, $z = 0.282$) and Abell 2218 ($\Delta T = 797 \mu\text{K}$, $z = 0.171$). The contour interval is $60 \mu\text{Jy}$ (left) and $80 \mu\text{Jy}$ (right). The dashed contours indicate a decrease in the received radio flux density. (Figure adapted from Jones et al., *MNRAS*, 357, 518, 2005.)

To incorporate this new feature, we introduce a two-component model of the universe, one that includes both the total density of matter (baryonic and dark), ρ_m , and the equivalent mass density of relativistic particles (such as neutrinos and CMB photons), ρ_{rel} . It is the equation of state $P = wu$ (Eq. 29.52) that determines whether we count a particle as matter ($w_m = 0$), a relativistic particle (a photon or neutrino, for which $w_{\text{rel}} = 1/3$), or (in Section 29.3) dark energy ($w_\Lambda = -1$). (The gravitational effect of the neutrinos' mass clearly persists; however, we will neglect the neutrinos' contribution to the value of $\Omega_{m,0}$ of roughly 0.003.)

Equation (29.53) shows that particles belonging to different categories are diluted differently by the expansion of the universe. Of course, at earlier epochs when the universe was much hotter, even massive particles were relativistic. For example, an electron gas at $T > 6 \times 10^9 \text{ K}$ has $kT > m_e c^2$, implying that the electron gas is relativistic and its equation of state is described by $w_{\text{rel}} = 1/3$ (see Problem 29.24). However, we will ignore such complications for the remainder of this chapter and will consider photons and neutrinos as our only relativistic particles.

With both matter and relativistic particles included, Eq. (29.10) becomes

$$\left[\left(\frac{1}{R} \frac{dR}{dt} \right)^2 - \frac{8}{3} \pi G (\rho_m + \rho_{\text{rel}}) \right] R^2 = -kc^2. \quad (29.65)$$

The equivalent mass density of the CMB photons comes from the energy density of black-body radiation, Eq. (9.7),

$$u_{\text{rad}} = aT^4, \quad (29.66)$$

where a is the radiation constant. We will rewrite this in the form

$$u_{\text{rad}} = \frac{1}{2} g_{\text{rad}} a T^4, \quad (29.67)$$

where g_{rad} is the number of **degrees of freedom** of a photon. The value of g reflects the number of spin states n_{spin} and the possible existence of an antiparticle ($n_{\text{anti}} = 1$ or 2). A photon is its own antiparticle ($n_{\text{anti}} = 1$) and can exist in $n_{\text{spin}} = 2$ spin states, corresponding to its two possible polarizations with its spin parallel or antiparallel to its motion. Thus

$$g_{\text{rad}} = 2 \quad (29.68)$$

for photons, as expected.

Neutrino Decoupling

We will neglect the small mass of the other relativistic particle we are considering, the neutrino, and treat it as a massless particle. The very early universe was sufficiently dense that neutrinos attained thermal equilibrium, with a spectrum very similar to that of blackbody radiation, Eq. (3.22), except the “ -1 ” in the denominator of that equation becomes a “ $+1$ ” for neutrinos. This occurs because photons are bosons, described by Bose–Einstein statistics, while neutrinos are fermions, described by Fermi–Dirac statistics; recall Section 10.2. Although the cosmic neutrino background has yet to be observed (not surprising given the challenge of detecting solar neutrinos), we have confidence that it exists.

Recall that there are three types (or flavors) of neutrinos—electron neutrinos, muon neutrinos, and tau neutrinos—and that each neutrino has a corresponding antineutrino. The total energy density of all three flavors is given by

$$u_\nu = 3 \times \frac{7}{8} \times a T_\nu^4 = 2.625 a T_\nu^4, \quad (29.69)$$

where the $7/8$ derives from the “ $+1$ ” in the expression for Fermi–Dirac statistics, and T_ν is the temperature of the neutrinos. As before, we write this as

$$u_\nu = \frac{1}{2} \left(\frac{7}{8} \right) g_\nu a T_\nu^4, \quad (29.70)$$

where

$$g_\nu = 6 \quad (29.71)$$

for neutrinos.

In general,

$$g = (\# \text{ types}) n_{\text{anti}} n_{\text{spin}}. \quad (29.72)$$

There is an antineutrino for each of the three types of neutrino so $n_{\text{anti}} = 2$, and neutrinos have one spin state (all neutrinos are left-handed) so $n_{\text{spin}} = 1$. We therefore recover $g_\nu = 3 \times 2 \times 1 = 6$.

The usual T in cosmology is always taken to be the temperature of the blackbody photons. However, the T_ν in Eq. (29.70) is the temperature of the neutrinos. For $T > 3.5 \times 10^{10}$ K, these temperatures are the same, and $T = T_\nu$. However, as the temperature dropped below about 3.5×10^{10} K, the expansion of the universe diluted the number density of neutrinos, and the neutrinos ceased to interact with other particles. Essentially, the cosmos expanded faster than the neutrino interaction rate, and the neutrinos decoupled from the other constituents of the universe. Since the time of **neutrino decoupling**, the neutrinos have expanded and cooled at their own rate, independently of the CMB.

The Energy Density of Relativistic Particles

Because the annihilation of electrons with positrons continued to supply energy to the photons (via $e^- + e^+ \rightarrow \gamma + \gamma$) but not to the neutrinos, the neutrino temperature is somewhat less than the temperature of the CMB photons. Although it is beyond the scope of this book, it can be shown that T_ν is related to the temperature T of the CMB photons by

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T. \quad (29.73)$$

The total neutrino energy density is therefore²⁰

$$u_\nu = \frac{1}{2} \left(\frac{7}{8}\right) g_\nu \left(\frac{4}{11}\right)^{4/3} aT^4 = 0.681 aT^4. \quad (29.74)$$

Thus the energy density for relativistic particles, both photons and neutrinos, is

$$u_{\text{rel}} = \frac{1}{2} g_* aT^4, \quad (29.75)$$

where

$$g_* = g_{\text{rad}} + \left(\frac{7}{8}\right) g_\nu \left(\frac{4}{11}\right)^{4/3} = 3.363 \quad (29.76)$$

is the **effective number of degrees of freedom** of the relativistic particles. We also define the equivalent mass density of relativistic particles as

$$\rho_{\text{rel}} = \frac{u_{\text{rel}}}{c^2} = \frac{g_* aT^4}{2c^2}. \quad (29.77)$$

This value of g_* is valid back to the end of electron–positron annihilation, at about $t = 1.3$ s. For the higher temperatures of the very early universe ($t < 1$ s) discussed in Chapter 30, however, we will encounter a greater number of relativistic particles, and the value of g_* will grow accordingly.

²⁰The factors of $7/8$ and $(4/11)^{4/3}$ do not describe degrees of freedom, and therefore we separate them from the definition of g_ν .

Employing Eq. (29.8), Eq. (29.65) becomes

$$H^2 [1 - (\Omega_m + \Omega_{\text{rel}})] R^2 = -kc^2, \quad (29.78)$$

where

$$\Omega_m = \frac{\rho_m}{\rho_c} = \frac{8\pi G\rho_m}{3H^2} \quad (29.79)$$

is the density parameter for matter (both baryonic and dark), and

$$\Omega_{\text{rel}} = \frac{\rho_{\text{rel}}}{\rho_c} = \frac{8\pi G\rho_{\text{rel}}}{3H^2} = \frac{4\pi Gg_*aT^4}{3H^2c^2} \quad (29.80)$$

is the density parameter for relativistic particles (both photons and neutrinos). Note that Eq. (29.78) implies that for a flat ($k = 0$) two-component universe, $\Omega_m + \Omega_{\text{rel}} = 1$. Inserting $T_0 = 2.725$ K, we find that

$$\Omega_{\text{rel},0} = 8.24 \times 10^{-5},$$

which is very small compared with $[\Omega_{m,0}]_{\text{WMAP}} = 0.27$.

Transition from the Radiation Era to the Matter Era

Recalling that $w_{\text{rel}} = 1/3$ for relativistic particles, Eq. (29.53) yields

$$R^4 \rho_{\text{rel}} = \rho_{\text{rel},0}, \quad (29.81)$$

which shows how the equivalent mass density of relativistic particles varies with the scale factor R . By comparing this with Eq. (29.5),

$$R^3 \rho_m = \rho_{m,0}, \quad (29.82)$$

for massive particles, we notice that ρ_{rel} increases more rapidly than the mass density ρ_m as the scale factor becomes smaller. As $R \rightarrow 0$ in the early universe, therefore, there must have been an early era when the radiation (i.e., all relativistic particles, not just photons and neutrinos) dominated and governed the expansion of the universe. The transition from this **radiation era** to the present **matter era** occurred when the scale factor satisfied $\rho_{\text{rel}} = \rho_m$, or $\Omega_{\text{rel}} = \Omega_m$. From Eqs. (29.79–29.82), the equality of Ω_{rel} and Ω_m occurred when the scale factor was

$$R_{r,m} = \frac{\Omega_{\text{rel},0}}{\Omega_{m,0}} = 4.16 \times 10^{-5} \Omega_{m,0}^{-1} h^{-2},$$

with a WMAP value of

$$R_{r,m} = 3.05 \times 10^{-4}.$$

This corresponds to a redshift (Eq. 29.4) of

$$z_{r,m} = \frac{1}{R_{r,m}} - 1 = 2.41 \times 10^4 \Omega_{m,0} h^2,$$

which for WMAP values is

$$z_{r,m} = 3270.$$

This is in very good agreement with the WMAP result,

$$[z_{r,m}]_{\text{WMAP}} = 3233^{+194}_{-210},$$

for the redshift when the the universe passed from being radiation-dominated to being matter-dominated.

Using $RT = T_0$ (Eq. 29.58), the temperature at this transition was

$$T_{r,m} = \frac{T_0}{R_{r,m}} = 6.56 \times 10^4 \Omega_{m,0} h^2 \text{ K},$$

or

$$T_{r,m} = 8920 \text{ K}$$

using WMAP values. Thus, when the universe had cooled to 8920 K and typical separations were some 4×10^{-5} of their present extent, relativistic particles ceased to govern the cosmic expansion, and matter assumed a dominant role.

Expansion in the Two-Component Model

We are now ready to determine how the early universe expanded with time. To discover how the scale factor, R , behaved during the radiation era, we begin by substituting Eqs. (29.81) and (29.82) into Eq. (29.65) to find

$$\left[\left(\frac{dR}{dt} \right)^2 - \frac{8}{3} \pi G \left(\frac{\rho_{m,0}}{R} + \frac{\rho_{\text{rel},0}}{R^2} \right) \right] = -kc^2. \quad (29.83)$$

Because the early universe was essentially flat, we can set $k = 0$ and use a bit of algebra to obtain

$$\int_0^R \frac{R' dR'}{\sqrt{\rho_{m,0} R' + \rho_{\text{rel},0}}} = \sqrt{\frac{8\pi G}{3}} \int_0^t dt'.$$

Integrating this eventually yields an expression for the age of the universe as a function of the scale factor R :

$$t(R) = \frac{2}{3} \frac{R_{r,m}^{3/2}}{H_0 \sqrt{\Omega_{m,0}}} \left[2 + \left(\frac{R}{R_{r,m}} - 2 \right) \sqrt{\frac{R}{R_{r,m}} + 1} \right], \quad (29.84)$$

where

$$\frac{2}{3} \frac{R_{r,m}^{3/2}}{H_0 \sqrt{\Omega_{m,0}}} = 5.51 \times 10^{10} h^{-4} \Omega_{m,0}^{-2} \text{ s} = 1.75 \times 10^3 h^{-4} \Omega_{m,0}^{-2} \text{ yr}.$$

The time $t_{r,m}$ of the transition from a radiation-dominated to a matter-dominated universe may be found by setting $R/R_{r,m} = 1$ and using WMAP values of $h = 0.71$ and $\Omega_{m,0} = 0.27$ to obtain

$$t_{r,m} = 1.74 \times 10^{12} \text{ s} = 5.52 \times 10^4 \text{ yr.} \quad (29.85)$$

The form of Eq. (29.84) becomes simpler deep in the radiation era, when $R \ll R_{r,m}$. It is left for you (Problem 29.27) to show that in this limit the factors of $\Omega_{m,0}$ cancel, resulting in

$$R(t) = \left(\frac{16\pi G g_* a}{3c^2} \right)^{1/4} T_0 t^{1/2} \quad (29.86)$$

$$= (1.51 \times 10^{-10} \text{ s}^{-1/2}) g_*^{1/4} t^{1/2}. \quad (29.87)$$

This shows that during the radiation era, $R \propto t^{1/2}$. Using $T = T_0/R$ quickly reveals the temperature deep in the radiation era:

$$T(t) = \left(\frac{3c^2}{16\pi G g_* a} \right)^{1/4} t^{-1/2} \quad (29.88)$$

$$= (1.81 \times 10^{10} \text{ K s}^{1/2}) g_*^{-1/4} t^{-1/2}. \quad (29.89)$$

At the other extreme, for $R \gg R_{r,m}$, Eq. (29.84) becomes

$$t(R) = \frac{2}{3} \frac{R^{3/2}}{H_0 \sqrt{\Omega_{m,0}}} \quad (29.90)$$

so

$$R(t) = \left(\frac{3}{2} H_0 t \sqrt{\Omega_{m,0}} \right)^{2/3} = \left(\frac{3\sqrt{\Omega_{m,0}}}{2} \right)^{2/3} \left(\frac{t}{t_H} \right)^{2/3}, \quad (29.91)$$

using $t_H = 1/H_0$ for the Hubble time. As expected, this displays the $R \propto t^{2/3}$ dependence we found earlier in Eq. (29.30) for a flat universe of pressureless dust. Equation (29.90) can be expressed in terms of z using $R = 1/(1+z)$ to obtain

$$\frac{t(z)}{t_H} = \frac{2}{3} \frac{1}{(1+z)^{3/2} \sqrt{\Omega_{m,0}}}, \quad (29.92)$$

which may be compared with Eq. (29.40) for a flat universe of pressureless dust. Evaluating these for $R = 1$ ($z = 0$) and using WMAP values gives the age of the universe as 17.7 billion years, more than six billion years more than the mean age of the oldest globular clusters (see page 1158). However, as we will see later, this estimate of the age of the universe is still about one billion years too short, as determined by a full analysis of the WMAP results.

Big Bang Nucleosynthesis

The process that manufactured the lightest elements in the early universe is known as **Big Bang nucleosynthesis**. We are now prepared to ask, why is approximately one-quarter of the mass of the universe in the form of helium? The temperature at time t during the radiation era is given by Eq. (29.89). At a temperature just below 10^{12} K ($t \sim 10^{-4}$ s), the universe contained a mixture of photons (γ), electron–positron pairs, and electron and muon neutrinos and their antiparticles ($\nu_e, \nu_\mu, \bar{\nu}_e, \bar{\nu}_\mu$). There were also a smaller number of protons and neutrons, about five for every 10^{10} photons, that were constantly being transformed into each other via the reactions

$$n \rightleftharpoons p^+ + e^- + \bar{\nu}_e \quad (29.93)$$

$$n + e^+ \rightleftharpoons p^+ + \bar{\nu}_e \quad (29.94)$$

$$n + \nu_e \rightleftharpoons p^+ + e^-. \quad (29.95)$$

These constant conversions were easily accomplished because the mass difference between a proton and a neutron is only

$$(m_n - m_p)c^2 = 1.293 \text{ MeV},$$

while the characteristic thermal energy of particles at 10^{12} K is $kT \simeq 86$ MeV. The Boltzmann equation (Eq. 8.6) gives the equilibrium ratio of the number density of neutrons, n_n , to the number density of protons, n_p , as²¹

$$\frac{n_n}{n_p} = e^{-(m_n - m_p)c^2/kT}. \quad (29.96)$$

At 10^{12} K, this ratio is 0.985. The numbers of neutrons and protons were nearly equal because the mass difference between the protons and neutrons is negligible at such a high temperature.

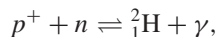
As the universe expanded and the temperature fell, the ratio of the number densities continued to be given by Eq. (29.96) as long as reactions (29.93–29.95) proceeded fast enough to reach equilibrium. Detailed calculations show, however, that when the temperature had declined to about 10^{10} K, the timescale for these reactions exceeded the characteristic timescale of the expansion given by $1/H(t) = 2t$; see Problem 29.26.

At a bit above 10^{10} K, the reaction rates decreased significantly, for two reasons. First, the expansion had reduced the energy of the neutrinos until they were unable to participate in reactions (29.93–29.95). Also, shortly thereafter, the characteristic thermal energy of the photons, kT , fell below the 1.022-MeV threshold for creating electron–positron pairs via the pair-production process $\gamma \rightarrow e^- + e^+$ (see Problem 29.30). As a result, the electrons and positrons annihilated each other without being replaced, leaving only a small remainder of excess electrons. For these reasons, the neutrons could not be replenished as fast as they were destroyed, and there was not enough time for these reactions to reach equilibrium. In a sense, the creation of new neutrons could not keep up with the rate of expansion

²¹The equal statistical weights of the neutron and proton cancel in Eq. (8.6).

of the universe. The ratio of the number densities then became “frozen” at its value of $n_n/n_p = 0.223$ when $T \simeq 10^{10}$ K.

At this point, there were 223 neutrons for every 1000 protons (or 446 neutrons for every 2000 protons, which will be used in what follows), and essentially no more neutrons were being created. The beta decay reaction [the forward reaction in Eq. (29.93)] continued to operate, however, converting neutrons into protons with a half-life of $\tau_{1/2} = 614$ s = 10.2 min. It was not yet possible for the protons and neutrons to combine to form deuterium nuclei (${}^2_1\text{H}$) via

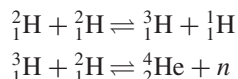


because at temperatures exceeding 10^9 K, the energetic radiation quickly dissociated the nuclei (recall page 1165). As a result, the neutrons and protons remained separated until the temperature had dropped from 10^{10} K to 10^9 K. According to Eq. (29.89) with $g_* = 3.363$, this took approximately

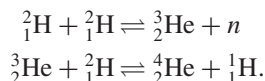
$$t(10^9 \text{ K}) - t(10^{10} \text{ K}) = 178 \text{ s} - 1.78 \text{ s} \approx 176 \text{ s}.$$

From the law of radioactive decay, Eq. (15.10), in this amount of time the 446 neutrons mentioned previously declined to 366, and the number of protons rose to 2080.

Below 10^9 K the neutrons and protons readily combined to form as many deuterium nuclei as possible. A number of reactions then led to the formation of ${}^4_2\text{He}$, the most tightly bound nucleus involved in Big Bang nucleosynthesis. The most efficient reactions leading to ${}^4_2\text{He}$ include



and



[Note that these reactions differ from those of the pp chain (Eqs. 10.37–10.45), which produce helium in the cores of stars.²²] No other nuclei were formed with abundances approaching that of ${}^4_2\text{He}$, although there were traces of ${}^2_1\text{H}$, ${}^3_2\text{He}$, and ${}^7_3\text{Li}$ (from the reaction ${}^4_2\text{He} + {}^3_1\text{H} \rightarrow {}^7_3\text{Li} + \gamma$). Figure 29.13 shows the network of reactions involved in Big Bang nucleosynthesis. Our sample of 366 neutrons and 2080 protons could form 183 ${}^4_2\text{He}$ nuclei, with 1714 protons (${}^1_1\text{H}$) left over. Because a ${}^4_2\text{He}$ nucleus is four times more massive than a ${}^1_1\text{H}$ nucleus, the preceding analysis shows that the mass fraction of ${}^4_2\text{He}$ in the universe should have been about

$$\frac{4(183)}{1714 + 4(183)} = 0.299.$$

²²The most obvious route to helium-4 is ${}^2_1\text{H} + {}^2_1\text{H} \rightleftharpoons {}^4_2\text{He} + \gamma$. However, this is a “forbidden” reaction and has a negligibly small cross section.

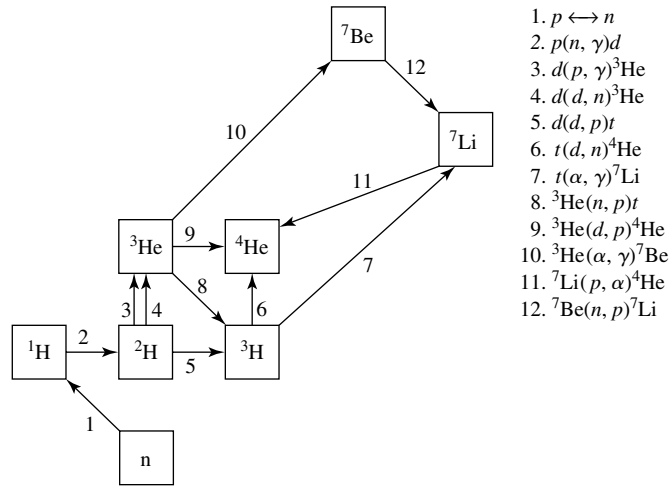


FIGURE 29.13 The reaction network that is responsible for Big Bang nucleosynthesis. The letter “d” stands for deuterium, and “t” stands for tritium. (Figure adapted from Nollett and Burles, *Phys. Rev. D*, 61, 123505, 2000.)

This rough estimate is consistent with the primordial percentage of helium inferred from observations, between 23% and 24%.

Because essentially all of the available neutrons were incorporated in the helium-4 (^4He) nuclei, the abundance of helium-4 was insensitive to the density of the universe at the time. However, the amounts of deuterium, helium-3, and lithium-7 manufactured in this way depend sensitively on the density of ordinary matter at the time of the reactions. Figure 29.14 shows the abundances of these nuclei as a function of the prevailing *present* density of baryonic matter. Comparing the theoretical curves with the observations makes it apparent that the present density of baryonic matter probably lies between 2 and $5 \times 10^{-28} \text{ kg m}^{-3}$, only a few percent of the critical density of $1.88 \times 10^{-26} h^2 \text{ kg m}^{-3}$. This explanation of the abundances of the light elements that were not manufactured by stars is one of the greatest achievements of the Big Bang theory.

The Origin of the Cosmic Microwave Background

Now that we have described the nature of the universal expansion, let’s return to the question of the origin of the CMB. When we observe the cosmic microwave background, what are we actually viewing? The copious electrons in the hot environment of the very early universe obstructed the photons of the cosmic microwave background, allowing them to travel only relatively short distances before being scattered. The scattering of photons by free electrons kept the electrons and photons in thermal equilibrium, meaning that they had the same temperature.²³ However, as the expansion of the universe diluted the number density of

²³The CMB photons are also scattered by free protons, but the cross section for photon–proton scattering is smaller than the Thomson cross section for electrons, Eq. (9.20), by a factor of m_e^2/m_p^2 , so it can be neglected. The Coulomb interaction between the electrons and protons kept the protons in thermal equilibrium with the electrons and photons.

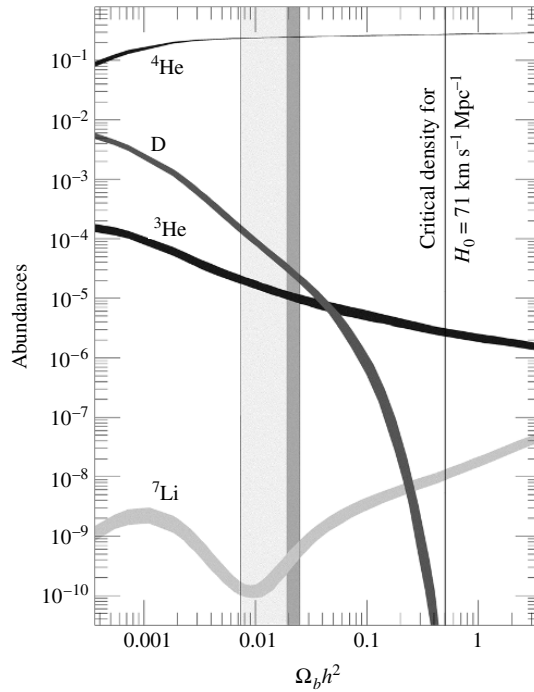


FIGURE 29.14 The calculated mass abundances of helium-4, deuterium, helium-3, and lithium-7 as a function of the present density of baryonic matter in the universe. The wide bar delineates the consistency interval, the range of values of $\Omega_{b,0}h^2$ that agree with the observed abundances. The narrow dark stripe at the right edge of the consistency interval corresponds to the abundances of primeval deuterium measured using the Lyman- α forest of absorption lines in high- z molecular clouds observed in front of quasars. The WMAP value of $\Omega_{b,0}h^2 = 0.0224$ runs down the center of the dark stripe, and the WMAP value of the critical density ($\Omega_{b,0}h^2 = 1h^2 = 0.504$) is shown at the right. Note that the agreement between the theoretical and observed abundances spans nine orders of magnitude. (Figure adapted from Schramm and Turner, *Rev. Mod. Phys.*, 70, 303, 1998.)

free electrons, the average time between scatterings of a photon by an electron gradually approached the characteristic timescale of the universal expansion,

$$\tau_{\text{exp}}(t) \equiv \left(\frac{1}{R(t)} \frac{dR(t)}{dt} \right)^{-1} = \frac{1}{H(t)}.$$

This expression is analogous to that of the pressure scale height, Eq. (10.69). As the time of **decoupling** approached, the photons became increasingly disengaged from the electrons.²⁴

If the electrons had remained free, decoupling would have occurred when the universe was about 20 million years old (see Problem 29.32). However, when the universe was only some one million years old (10^{13} s), another important event altered the opacity of the

²⁴You may recall the discussion of neutrino decoupling on page 1173.